Mission Antyodaya 2020 Village Cluster Analysis

Profiling rural development using 64 selected survey features and K-means clustering across 21 categories.

REPORT PREPARED BY

Amit Kumar

CoRE Stack

SUPERVISED BY

Prof. Aaditeshwar Seth

IIT Delhi

A village-level analysis of 21 development categories using Mission Antyodaya 2020 data, with Low, Medium, and High cluster classifications for rural planning and prioritization.

July 2026

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Acknowledgement

Ananya Singh, IIT Delhi, helped with data preparation and analysis.

1 Introduction

This report presents a village-level cluster analysis of the Mission Antyodaya 2020 dataset. The objective is to convert a large, multi-sector rural development survey into an interpretable set of category and feature indicators that can support planning, prioritization, and spatial comparison across villages in India.

The analysis organizes selected Mission Antyodaya variables into 64 feature indicators and 21 broader development categories. Each category captures a different aspect of rural development, including institutional capacity, social protection, infrastructure, financial inclusion, agriculture, livelihoods, health, education, water, sanitation, and connectivity. Villages are classified into Low, Medium, and High clusters where the underlying data have sufficient variation; binary indicators are classified into Low and High groups.

The updated output system is designed for both analytical and report-facing use. The village-level analytical source retains category clusters and values, 64 normalized feature values, and the raw survey fields used to construct them. It is joined with CoRE Stack administrative-boundary layers to create the pan-India GeoPackage and GeoServer layer. No output publishes a `*_feat_cluster` column. The generated Excel sheet and village Know Your Landscape output use a smaller contract: 21 category clusters, 21 category values, and 109 raw survey fields, with no feature columns. Together, these outputs support village-level mapping, comparison, reporting, and exploration through the Antyodaya 2020 Explorer.

2 Data Summary

The analysis uses state-wise Mission Antyodaya 2020 raw CSV files as the primary input. After deduplication and village-level aggregation, the working dataset contains more than 6.3 lakh village records. The selected survey variables are transformed into normalized indicators and then aggregated into development categories for cluster classification.

Table 1: Data summary

Metric	Value
Raw state CSV files	32
Raw rows	648,358
Deduplicated village rows	631,249
Duplicate village groups	12,618
Duplicate extra rows aggregated	17,109
Raw parameters in headers	198
Raw parameters used by current mapping	121
Feature indicators created	64
Development categories created	21

3 Data Processing and Outputs

The processing workflow standardizes raw Mission Antyodaya survey data, constructs feature indicators, aggregates them into development categories, and prepares both tabular and geospatial outputs. The final files are available in the project Google Drive folder.

3.1 Processing workflow

- State-wise raw CSV files are loaded with only the columns required by the variable-mapping configuration.
- Village identity is grouped by state, district, sub-district, and village code, while readable state, district, sub-district, and village names are retained for interpretation.
- Raw survey parameters are converted into feature indicators through ratio, binary-presence, ordinal-score, and composite-score calculations.
- During construction, feature indicators are normalized and grouped into Low, Medium, and High bands where the metric has enough variation; binary-only indicators use Low/High. These intermediate feature bands are used to build the category score but are not published as output columns.
- Development categories are scored from these internal feature bands and clustered into final Low, Medium, and High or Low/High labels.
- Final analytical outputs are consolidated into a standardized village-level CSV containing category cluster/value columns, normalized feature-value columns, and the raw survey fields.
- The consolidated village-level CSV is joined with CoRE Stack administrative-boundary layers to create a pan-India geospatial asset for mapping and spatial analysis.
- The Excel and village Know Your Landscape flows select category cluster/value columns and raw survey fields from the published layer; they do not carry feature columns forward.

3.2 Consolidated outputs

The 2020 analytical output is kept as a consolidated village-level CSV and an indexed SQLite sidecar under `data/base_resources/`. The source contains the Antyodaya 2020 cluster analysis for villages across India. Its main naming conventions are:

- `*_cat_cluster`: category-level cluster classification.
- `*_cat_value`: normalized category-level value.
- `*_feat_value`: normalized feature-level indicator value.

- raw survey names such as `availability_of_piped_tap_water`: the reported Mission Antyodaya value used by one or more categories.

There is no `*_feat_cluster` field in the analytical CSV, GeoPackage, GeoServer layer, Excel sheet, or Know Your Landscape output. Read a result in this order: begin with the `*_cat_cluster`, check its `*_cat_value`, and then use the normalized `*_feat_value` fields in the analytical spatial layer to understand the contribution of each component. The report-facing Excel and Know Your Landscape contract keeps the category cluster/value pairs and 109 raw fields, but excludes feature fields.

The raw survey fields are included for traceability. They show the reported entries from which the normalized features were derived, but they do not share a common scale and are not analysis-ready. They must not be used to rank villages, define cross-category patterns, or replace the normalized category and feature values.

3.3 File inventory and access

Table 2: Generated analysis outputs

Output	Description
`data/base_resources/cs_antyodaya_2020_cluster_analysis.csv`	Consolidated village-level source for the local pipeline. The file contains category clusters and values, normalized feature values, and raw survey fields. It contains no feature-cluster columns. Its generated SQLite sidecar supports keyed runtime reads.
Published GeoPackage and GeoServer layer	Requested village geometries joined with the complete analytical attribute set. These outputs support spatial inspection, mapping, and regional comparison. They retain normalized feature values but no feature-cluster columns.
Generated Excel and village KYL output	Report-facing fields containing the 21 category clusters, 21 category values, and 109 raw survey fields; feature columns are not included.
antyodaya_variable_mapping.json	Variable-mapping configuration that defines how raw Mission Antyodaya survey parameters are transformed, weighted, normalized, aggregated into features and categories, and classified into final development clusters.
mission_antyodaya_gp_questionnaire_2020.pdf	Original Mission Antyodaya 2020 Gram Panchayat survey questionnaire used as the source reference for interpreting raw survey variables and feature definitions.
raw_files/{state}.csv	State-wise raw Mission Antyodaya 2020 CSV files used as the input data before feature construction, category aggregation, village-level consolidation, and geospatial joining.

Table 2: Generated analysis outputs

Output	Description
Antyodaya 2020 Explorer	Google Earth Engine application for exploring the pan-India Antyodaya 2020 analysis dataset, visualizing development indicators, and comparing village-level disparities across India: open the Antyodaya 2020 Explorer.

The Explorer provides an interactive entry point into the same analysis outputs. It can be used to view pan-India village-level patterns, compare development indicators spatially, and identify disparities across rural regions using the consolidated Antyodaya 2020 data products.

4 Observations

The cluster results show that rural development status is not moving uniformly across sectors. Some foundational services have relatively wider reach, especially **Maternal and Child Health**, **Road Connectivity**, **Agriculture Land Cultivation**, and parts of **Energy Access**. At the same time, several second-order systems remain weak: **Financial Inclusion**, **Agricultural Markets**, **Agriculture Support Services**, **Institutionalization**, **Water and Sanitation**, and most livelihood diversification categories. This suggests a common pattern: many villages may have basic connectivity, some public service access, and agricultural land use, but still lack the institutional, financial, market, and enterprise systems needed to convert infrastructure into durable livelihood gains.

The most important interpretation is that the High/Medium/Low clusters should not be read as simple "developed versus undeveloped" labels. Some categories represent universal needs, such as water, sanitation, financial inclusion, civic infrastructure, and social protection. Others are context-specific, such as forest livelihoods, fisheries, common pastures, alternative farming, and cottage industries. A Low score in forest resources or fisheries may sometimes indicate absence of ecological suitability rather than a development failure. Therefore, for action planning, the categories should be divided into two groups: **universal service deficits** and **livelihood opportunity specializations**.

A strong cross-cutting pattern is the gap between **physical access** and **productive conversion**. Road connectivity is comparatively strong, with only 14.3% villages in Low and 42.0% in High, but agricultural markets remain weak, with 71.2% villages in Low. Similarly, agriculture land cultivation is broadly strong, with only 6.2% Low and 40.1% High, but agriculture support services have only 10.5% High, and organic farming only 5.3% High. This means many villages are agricultural, connected, and potentially reachable, but lack the market access, storage, input support, risk support, soil testing, irrigation modernization, FPO/PACS strength, and credit linkage required for higher-value rural production.

Another major pattern is the gap between **household-level scheme reach** and **local institutional capacity**. Social protection has a relatively balanced distribution, but institutionalization, civic infrastructure, and financial inclusion remain weaker. This matters because delivery of pensions, PDS, credit, PMAY, Ujjwala, health entitlements, farmer insurance, and livelihood schemes depends not only on formal eligibility but also on local facilitation capacity. Villages with weak Panchayat infrastructure, weak public information systems, low SHG penetration, low bank/BC access, and low representative training may be less able to convert schemes into actual household benefits.

The results also imply that many interventions should not be category-wise silos. For example, a village with high farm employment but low irrigation, low agricultural support, low market access, and low institutionalization needs an integrated **productive agriculture package**, not separate isolated interventions. A village with high SHG federation but low SHG credit, low FPO/PACS, and low market access needs a **collective-to-enterprise transition package**. A village with good roads and electricity but low cottage industry, low financial inclusion, and weak markets needs an **enterprise activation package**.

5 Analysis Results

This section reads final category classes and feature distributions together. Each category plot shows feature values inside the final category class; for two-class categories, only Low and High panels are shown.

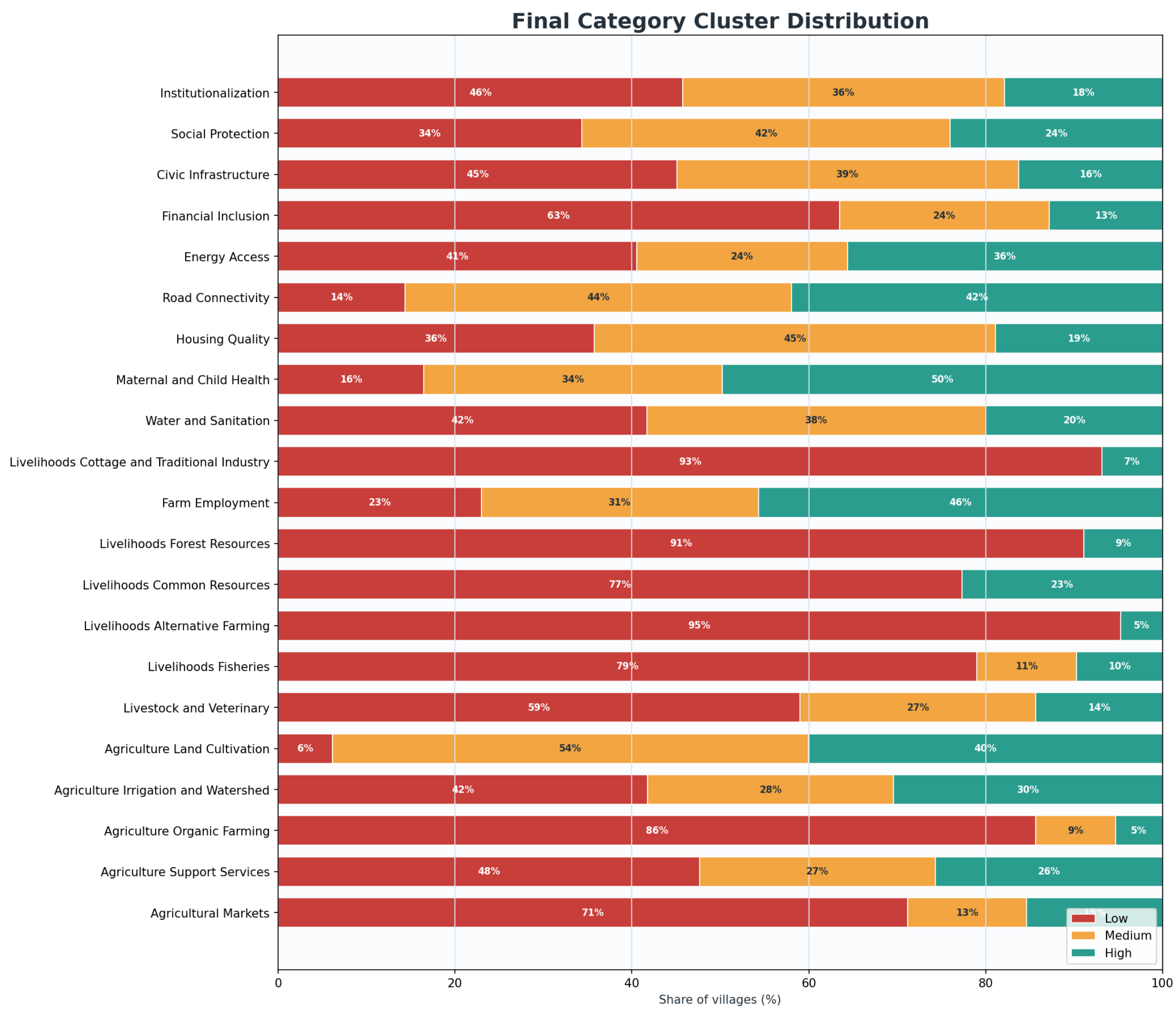

Figure 1: *category cluster distribution across all 21 categories*

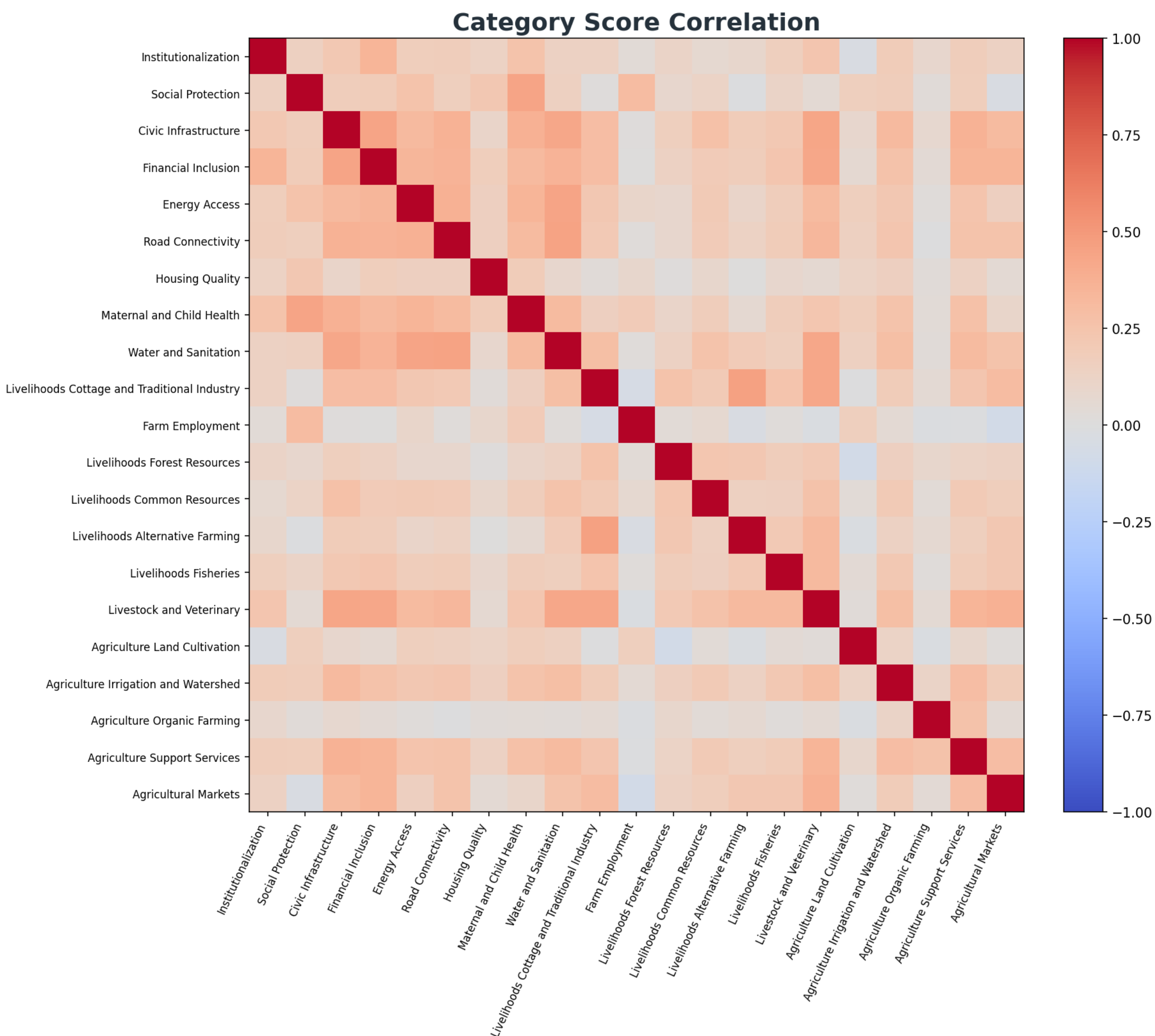

Figure 2: *correlation between normalised category scores*

5.1 Pan-India village Water and Sanitation patterns

The Water and Sanitation map reveals concentrations that cannot be read from a national table. It shows broad areas where neighbouring villages share stronger conditions, persistent lower-value belts, and sharp local breaks where weaker pockets sit within otherwise stronger regions. This makes it possible to see patterns that cross administrative boundaries as well as variation within the same state.

The same approach can be used with any of the 21 category values in the Antyodaya dataset. A reader can locate adjoining village groups, open the pattern through its normalised component features, and see whether water supply, sanitation, drainage, waste disposal, or recycling is shaping the result. The map can also be combined with maternal and child health, nutrition, water availability, or facilities access to identify concentrated disadvantages, compare possible intervention areas, and find stronger village clusters for field learning.

The 545,874 mapped village values produce the reported percentiles directly: 0.200 at the 10th percentile, 0.400 at the median, and 0.700 at the 90th percentile. These values fall on clean tenths because the category value is the equal-weight mean of five component class scores, where Low is 0, Medium is 0.5, and High is 1.

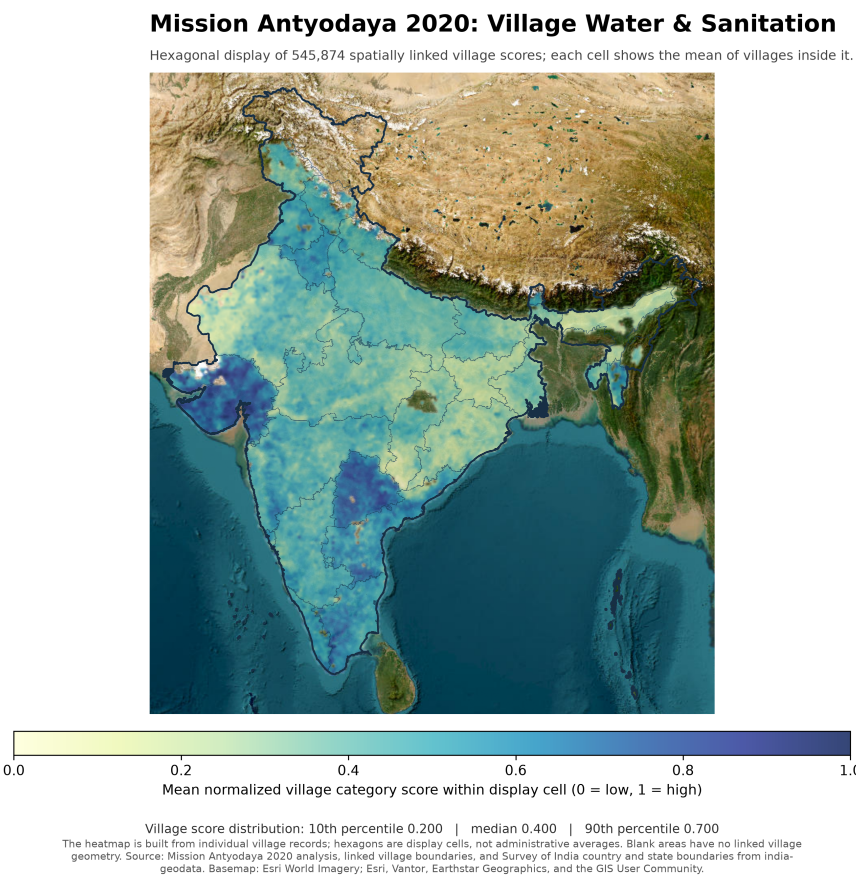

Figure 3: Village Water and Sanitation patterns across India. Darker areas show stronger category values and pale areas show weaker values.

6 Category Feature Box Plots

6.1 Institutionalization

The **Institutionalization** category measures the strength of village-level collective institutions. It is built from four features:

- **SHG Penetration** = households mobilized into SHGs divided by total households.
- **SHG Federation** = number of SHGs promoted or federated divided by total SHGs.
- **PG Penetration** = households mobilized into Producer Groups divided by total households.
- **FPO/PACS Strength** = score based on availability of Farmers Collective (Farmer Producer Organizations (FPOs)/Primary Agricultural Credit Societies (PACS)), where stronger presence of formal farmer/producer institutions receives a higher score.

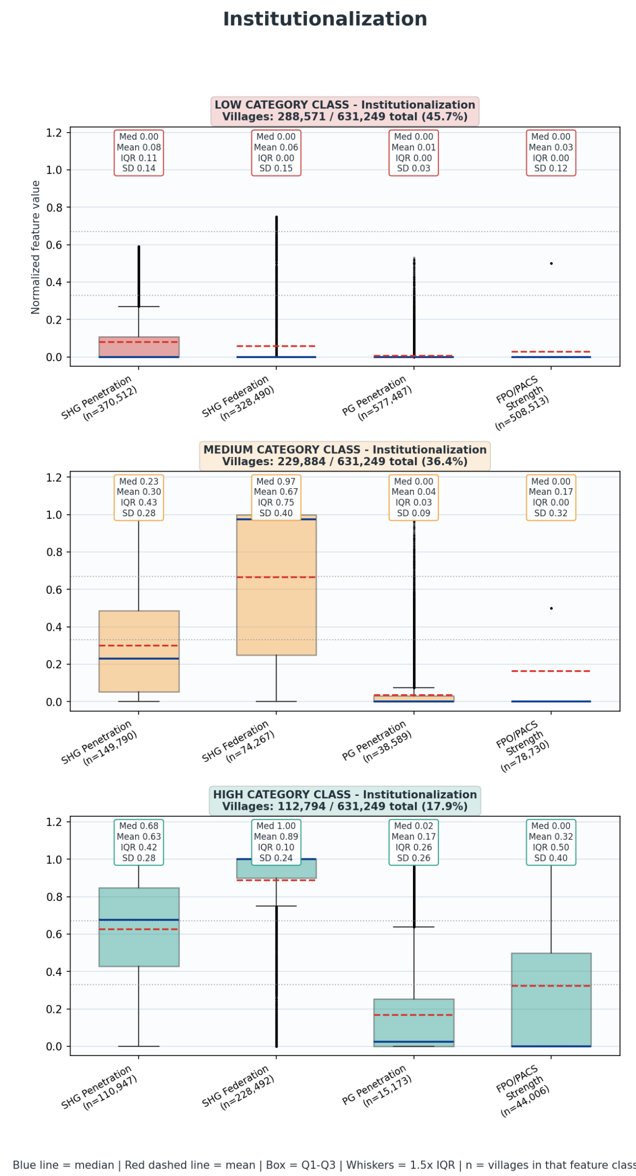

Figure 4: Institutionalization

The category captures both community-based social mobilization and producer-oriented collective organization. SHGs and SHG federations represent household and community mobilization, often with a strong women's collective dimension. Producer Groups, FPOs, and PACS represent a more market-facing and production-linked institutional layer. This category should be treated as an **enabling category**. It does not directly measure roads, water, income, health, or farm productivity. Instead, it measures whether the village has the collective structures required to plan, access schemes, aggregate demand, organize credit, support livelihoods, and negotiate with markets or government systems.

LOW Cluster Villages

Low Institutionalization villages are the most institutionally fragile group. These villages likely have weak SHG mobilization, weak federation structures, almost no Producer Group penetration, and limited FPO/PACS presence. It impacts development system very negatively. Villages with weak institutions may struggle to: identify eligible beneficiaries, organize women, farmers, or vulnerable households; access credit through SHGs or banks; form producer groups; aggregate agricultural produce; participate in government schemes; manage common assets; sustain livelihood interventions; negotiate with markets; maintain community infrastructure; support collective monitoring of service delivery.

LOW Institutionalization becomes especially concerning when it overlaps with low **Financial Inclusion**, low **Civic Infrastructure**, low **Social Protection**, low **Agriculture Support Services**, or low **Agricultural Markets**. In such villages, households may not only lack services; they may also lack the foundational mechanisms needed to demand, access, or sustain those services. For Low villages, the first priority should be **basic institutional formation and activation**. This includes SHG mobilization, SHG quality strengthening, federation formation where appropriate, Panchayat linkage, public information systems, and financial access. It may be premature to immediately push complex FPO or market-facing models in these villages unless the basic mobilization layer is first established.

MEDIUM Institutionalization villages

Medium Institutionalization villages are the most strategically important transition group. They account for **36.4%** of villages. These villages likely have some functioning SHG or federation base, but weak PG penetration and weak FPO/PACS strength. This group represents the clearest opportunity for institutional upgrading. The social base exists, but it has not yet been converted into a stronger economic or producer-facing system. In many Medium villages, SHGs may be active for savings, credit, or scheme participation, but not yet linked to producer groups, value chains, farm aggregation, local enterprise, or market access.

Actionably, medium villages should not be treated as institutionally weak in the same way as Low villages. They already have partial capacity. The planning focus should be to graduate from social mobilization to economic organization, credit linkage, producer aggregation, and convergence. The key question for Medium villages is: **how can existing SHG/federation capacity be converted into livelihood and development capacity?** Potential interventions may include: strengthening SHG bookkeeping and regularity; improving SHG-bank linkage; expanding SHG credit access; forming livelihood sub-groups within SHGs; creating Producer Groups from existing SHG or farmer networks; linking SHG federations with agriculture, livestock, fisheries, cottage industry, or nutrition interventions; identifying potential FPO/PACS linkages; using SHG federations for PMMVY, PMJAY, pension, NFSA, and other entitlement support; training community cadres for service delivery and monitoring.

HIGH Institutionalization villages

High Institutionalization villages form **17.9%** of the total. These villages have the strongest overall institutional base, especially in SHG penetration and SHG federation. However, the feature pattern shows that even High villages may not have strong PG or FPO/PACS systems. Therefore, the High category should be interpreted as **relative institutional strength**, not full institutional maturity. High Institutionalization villages are strong candidates for advanced livelihood and convergence interventions. Where SHGs and federations are already active, programmes can move beyond basic mobilization toward credit expansion, producer aggregation, women-led enterprise, agricultural value chains, dairy/fisheries groups, organic farming clusters, nutrition campaigns, social protection saturation, and local monitoring. Second, High villages can act as **institutional hubs** for nearby Low and Medium villages. They may provide demonstration value, peer learning, federation support, training support, or cluster-level aggregation.

6.2 Social Protection

Social Protection has a relatively balanced but still uneven distribution: **216,650 villages in Low, 262,915 in Medium, and 151,684 in High**, equivalent to **34.3% Low, 41.6% Medium, and 24.0% High**. This means the largest group of villages falls in the Medium category, suggesting that social protection systems are present in many places but are not uniformly saturated.

PDS Utilization is the strongest feature. In the internal bands used during the original category construction, 363,214 villages were High, compared with 233,274 Low and 34,761 Medium. These bands describe the analysis stage; they are not published feature-cluster fields. This implies that where NFSA eligibility is recorded, the flow of food grains through Fair Price Shops appears comparatively strong in many villages. However, this strength does not automatically translate into a high overall Social Protection category because the other features, especially pension coverage, remain weaker. **Pension Coverage is the weakest part of the category**, with **353,875 villages in Low**, 205,745 in Medium, and only 71,629 in High. This is important because pension benefits usually target older persons, widows, persons with disabilities, and other vulnerable groups. Low pension coverage can therefore indicate a deeper last-mile exclusion problem even in villages where PDS delivery is functioning. BPL and NFSA coverages indicate that vulnerability identification itself is uneven. Some villages may have high PDS utilization among the eligible population but low recorded eligibility relative to total households, while others may have high eligibility but incomplete benefit uptake.

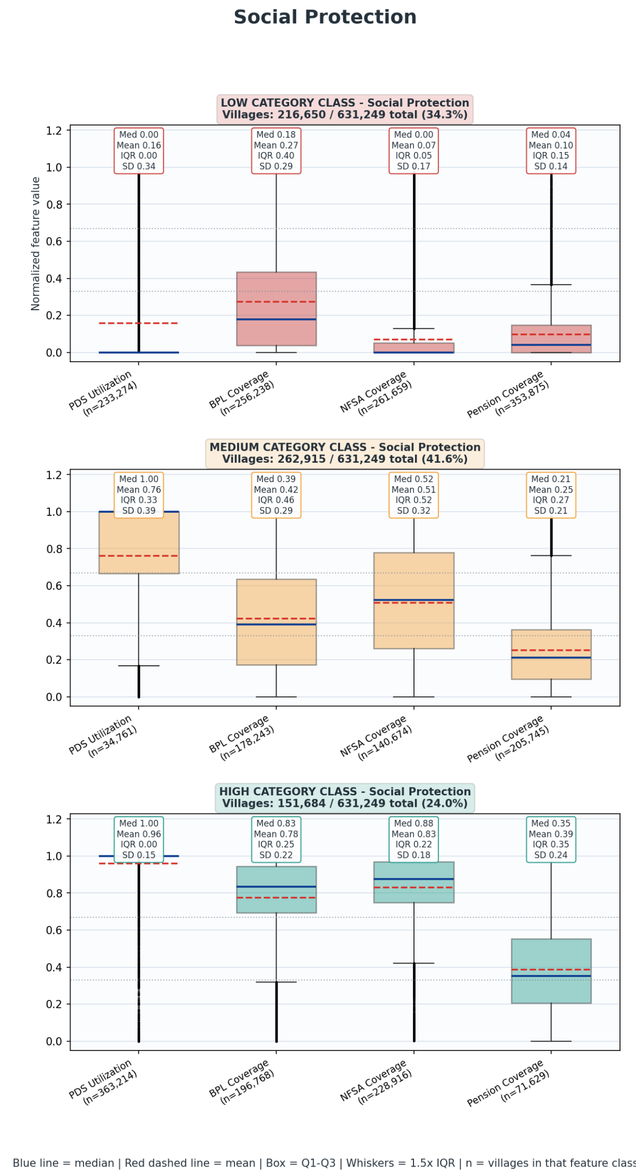

Figure 5: Social Protection

LOW: Under Socially Protected cluster

Low Social Protection villages are likely to have weak entitlement saturation across multiple welfare channels. These villages may have low NFSA eligibility recording, low BPL card coverage, low pension uptake, or a combination of these. A Low category classification should be interpreted as a potential welfare access risk, not just a poverty signal. The most concerning Low villages are those that also score low in **Financial Inclusion**, **Civic Infrastructure**, **Housing Quality**, and **Water and Sanitation**. In such villages, household deprivation and administrative exclusion may reinforce each other. Poor households may exist, but may not be fully documented, may lack banking access, may not receive scheme information, or may face weak Panchayat-level facilitation. Low Social Protection villages should therefore be prioritized for **entitlement verification camps**, **household-level eligibility**

audits, **pension saturation drives**, **ration-card correction**, and **NFSA beneficiary review**.

MEDIUM Social Protection villages

The Medium class is the largest group, which makes it the most important from a scale perspective. These villages likely have functioning components of the social protection system but incomplete convergence. For example, PDS delivery may be working, but pension coverage may be low; or NFSA eligibility may be moderate, but BPL coverage may not fully reflect household vulnerability. Medium Social Protection villages represent the "near-saturation" opportunity. They may not require the same foundational intervention as Low villages, but they need targeted gap closure. Villages should be profiled by the weakest feature: PDS utilization, BPL coverage, NFSA coverage, or pension coverage. This allows precise intervention instead of generic welfare drives. Medium villages should also be cross-analysed with **Maternal and Child Health**. If MCH is high but Social Protection is medium or low, it suggests that health-service outreach may be stronger than household welfare support.

HIGH Social Protection villages

High Social Protection villages show relatively stronger welfare delivery and entitlement coverage. But High should not be interpreted as "no poverty" or "no vulnerability." It means the measured welfare channels are functioning better relative to other villages. These villages can serve as reference cases for understanding delivery mechanisms. If a High Social Protection village also has strong Civic Infrastructure, Financial Inclusion, and Institutionalization, it may indicate that local governance and community institutions are helping scheme delivery. If a High Social Protection village has weak Financial Inclusion, it may mean food-grain delivery is functioning but cash-transfer or pension-related financial access may still be constrained. High Social Protection villages should be studied not only for welfare saturation, but also for whether welfare coverage is translating into better household outcomes. Cross-category checks with **Housing Quality**, **Water and Sanitation**, **MCH**, and **Livelihoods** can show whether social protection is acting as a stabilizer or whether it remains a standalone benefit channel.

6.3 Civic Infrastructure

Civic Infrastructure has **284,571 villages in Low, 243,967 in Medium, and 102,711 in High**, equivalent to **45.1% Low, 38.6% Medium, and 16.3% High**. This means only about one-sixth of villages fall in the High class, while nearly half remain Low. The feature-level pattern shows that **Panchayat Bhawan**, **Public Information Board**, and **Representative Orientation** have meaningful variation and appear to drive much of the category spread. During category construction, Panchayat Bhawan placed 262,001 villages in its internal High band and 369,248 in Low. Public Information Board has 257,782 High but 373,467 Low. Representative Orientation has 220,912 High, 46,205 Medium, and 364,132 Low. This suggests a large number of villages lack either physical Panchayat infrastructure, visible public information systems, or oriented representatives.

Public Library Availability is extremely weak, with 573,114 villages in Low and only 58,135 High. This indicates that library infrastructure is rare and should not dominate the interpretation of the category, but it remains important as a marker of civic knowledge infrastructure. Post Office Availability is also limited, with 492,650 villages in Low and 138,599 High. This matters because post offices can support communication, savings, payments, documentation, and last-mile service access, especially in areas where bank presence is weak. Representative Training and Orientation show large Low counts but also meaningful High counts. Training has 405,768 Low and 177,682 High; orientation has 364,132 Low and 220,912 High. Orientation appears more widespread than refresher training. This may suggest that initial orientation is more common than deeper or repeated capacity-building.

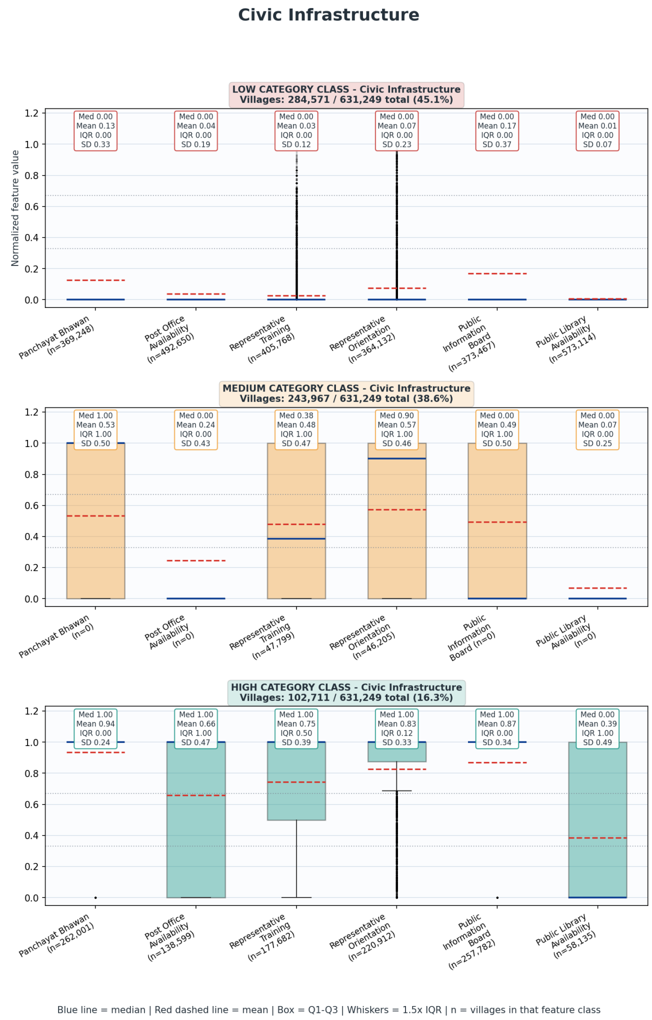

Figure 6: Civic Infrastructure

LOW Civic Infrastructure

Low Civic Infrastructure villages may lack the basic governance platform required for scheme convergence. If a village does not have a Panchayat Bhawan, updated information board, trained representatives, post office access, or public library, it may face weak administrative visibility and weaker public communication. These villages are likely to struggle in areas that depend on local facilitation: Social Protection, Financial Inclusion, Housing, Water and Sanitation, Agriculture Support Services, and Institutionalization. **For example, low civic infrastructure can reduce awareness of pension eligibility, NFSA lists, PMAY wait-list status, MGNREGA works, agricultural schemes, insurance enrolment, soil testing campaigns, and SHG/FPO support.** Low Civic Infrastructure should therefore be interpreted as an enabling-system deficit. It may not directly measure household welfare, but it affects whether households can access schemes, participate in planning, raise grievances, and receive timely information.

MEDIUM cluster

Medium Civic Infrastructure villages likely have partial governance capacity. They may not require heavy infrastructure investment in every case. Sometimes the missing element may be an updated public information board, refresher training of elected representatives, use of Panchayat Bhawan for service camps, or linkage with post-office/banking services. With modest investments, these villages may move into the High category and become effective convergence nodes for multiple development programmes.

HIGH Civic Infrastructure Villages

High Civic Infrastructure villages are likely to have stronger local administrative platforms. High Civic Infrastructure should be used as an implementation advantage: these villages can support more sophisticated planning, data validation, service camps, public disclosure, and convergence. **High Civic Infrastructure villages can also serve as administrative hubs for nearby villages, especially where road connectivity is good.** They may be suitable locations for block-level camps, SHG federation meetings, farmer training, social audit facilitation, digital service points, and grievance redressal days.

Interaction with other categories

Civic Infrastructure is one of the strongest enabling categories in the entire framework. It interacts with almost every other category because many village outcomes require local coordination. With **Social Protection**, civic infrastructure supports beneficiary identification, public disclosure, grievance handling, and list correction. Low Social Protection plus low Civic Infrastructure is a serious exclusion-risk combination. With **Financial Inclusion**, Panchayat spaces and public information systems can host banking camps, Jan Dhan activation drives, SHG credit camps, pension account correction, and bank correspondent awareness. Where Financial Inclusion is low but Civic Infrastructure is medium or high, the governance platform already exists and can be used to improve banking access. With **Institutionalization**, civic infrastructure can support SHG federation meetings, producer group formation, public meetings, and convergence with Panchayat planning. Strong civic infrastructure but weak institutionalization suggests that formal governance exists but community collectives are underdeveloped. With **Agriculture Support Services**, trained representatives and Panchayat platforms can help organize soil-testing camps, PMFBY/PMKPY enrolment, farmer advisory, input distribution, and watershed planning. This is especially important in villages with high land cultivation but weak agricultural services. With **Water and Sanitation**, public information boards and Panchayat-level planning can improve monitoring of piped water, drainage, sanitation, waste disposal, and community assets.

6.4 Financial Inclusion

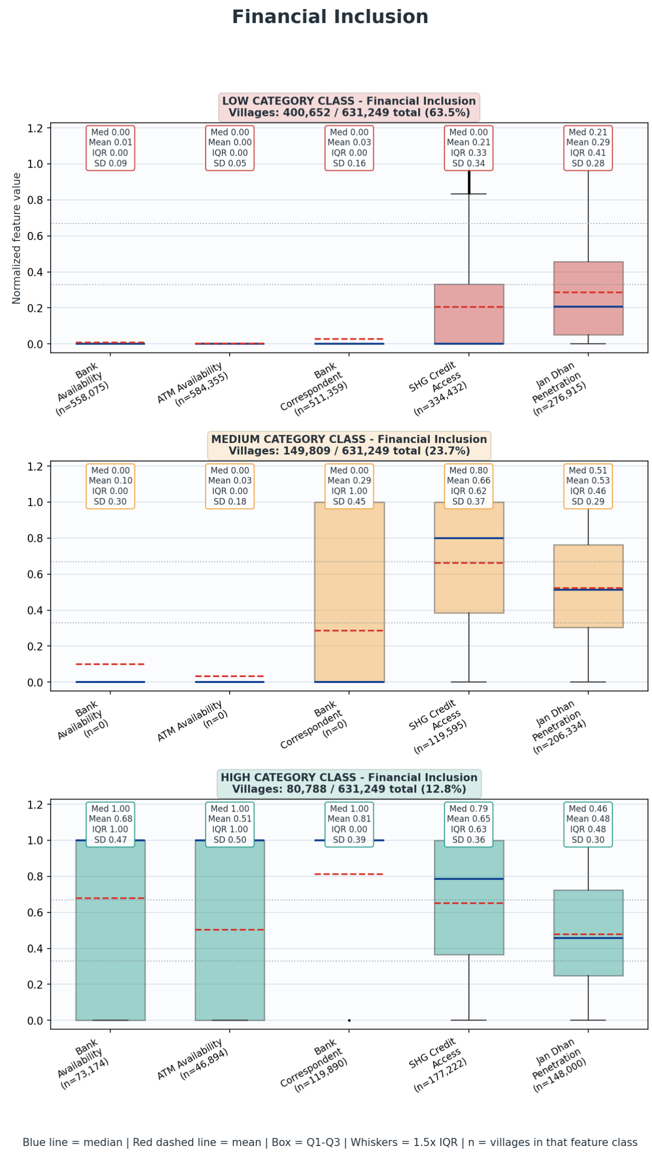

Figure 7: Financial Inclusion

Financial Inclusion is one of the weakest categories in the entire analysis. It has **400,652 villages in Low, 149,809 in Medium, and only 80,788 in High**, equivalent to **63.5% Low, 23.7% Medium, and 12.8% High**. The feature-level pattern explains why the category is so weak. It comprises of direct availability of different financial resources, such as availability of bank within the village, which would be low, since neighbouring villages are generally served by one nearby bank. **Bank Availability is low in 558,075 villages** and high in only 73,174. **ATM Availability is even weaker**, with 584,355 Low and 46,894 High. **Bank Correspondent availability is somewhat better**, but still limited: 511,359 Low and 119,890 High. This indicates that direct village-level financial infrastructure is sparse. The usage-based features are stronger than physical infrastructure but still uneven. **SHG Credit Access** has 334,432 Low, 119,595 Medium, and 177,222 High. **Jan Dhan Penetration** has 276,915 Low, 206,334 Medium, and 148,000 High. This suggests that household account coverage and SHG credit linkage exist in many places, but not enough to compensate for weak local service access. A key interpretation is that Financial Inclusion is not just about opening bank accounts. The cluster results show a layered deficit: physical banking infrastructure is limited, Bank Correspondent access is incomplete, and credit linkages through SHGs are uneven.

LOW Financial Inclusion

Low Financial Inclusion villages are likely to have weak access to formal financial services. They may lack a bank branch, ATM, internet-enabled business correspondent, household account penetration, or SHG credit linkage. Low Financial Inclusion --when combined with Low Social Protection becomes much more serious. Pension payments, DBT flows, welfare withdrawals, and benefit access may be constrained if the village lacks service points or functioning accounts. It is also serious when combined with Low Institutionalization, because SHGs and federations often act as bridges to credit and banking. A Low Financial Inclusion village with high Farm Employment and high Agriculture Land Cultivation is also a credit-risk zone. Farmers may be economically active but unable to access formal credit, crop insurance, input finance, or investment loans. **This can push households toward informal borrowing** and limit productivity improvements.

MEDIUM Financial Inclusion cluster

They may have Jan Dhan accounts but weak banking infrastructure, or SHG credit access without strong bank/ATM/BC availability. These villages represent the most practical upgrade opportunity. Medium villages should be targeted for service-density improvement: Bank Correspondent points, digital connectivity, SHG-bank linkage camps, PMJDY account activation, pension account correction, and credit counselling.

HIGH Financial Inclusion

High Financial Inclusion villages have stronger financial access and usage. These villages may have better bank/BC infrastructure, higher Jan Dhan coverage, and better SHG credit linkage. They can support higher-order development interventions such as enterprise finance, FPO credit, dairy credit, crop investment, insurance adoption, and market-linked production. High Financial Inclusion villages can become financial-service hubs for surrounding areas, particularly if they also have strong Road Connectivity and Civic Infrastructure. However, it doesn't guarantee that access to finance can be the final target. If a village has high Financial Inclusion but low Institutionalization, credit may be available but collective absorption capacity may be weak. If it has high Financial Inclusion but low Agricultural Markets, farmers may be able to borrow but may still struggle to sell produce at good prices. If it has high Financial Inclusion but low Agriculture Support Services, loans may not translate into productivity because technical and risk support are missing. The priority should become productive finance.

Cross Category Connections

Financial Inclusion has especially strong interaction with **Institutionalization**. SHGs, SHG federations, producer groups, and FPO/PACS structures can convert financial access into collective livelihood action. Where Institutionalization is medium or high but Financial Inclusion is low, the missing piece is credit and banking linkage. Where Financial Inclusion is high but Institutionalization is low, individual accounts may exist but collective economic organization is weak. It also interacts deeply with **Social Protection**. Welfare benefits are increasingly linked to bank accounts and digital payment systems. Low Financial Inclusion can weaken pension delivery, DBT access, and household ability to use entitlements. Therefore, villages with low Social Protection and low Financial Inclusion should be prioritized for combined welfare-finance camps. With **Civic Infrastructure**, financial inclusion can be improved through local platforms. Panchayat Bhawans and public information boards can be used for bank camps, Jan Dhan account correction, BC awareness, SHG credit drives, and pension payment facilitation. Strong Civic Infrastructure but weak Financial Inclusion is an actionable gap: the administrative platform exists, but financial access needs to be brought into it. With **Agriculture Land Cultivation**, **Agriculture Support Services**, and **Agricultural Markets**, financial inclusion determines whether farmers can invest in seeds, fertilizer, irrigation, mechanization, insurance, storage, and transport. If agriculture is active but financial inclusion is low, productivity may remain low because farmers cannot finance improvements. With **Livelihoods Cottage and Traditional Industry**, **Fisheries**, **Livestock and Veterinary**, and **Alternative Farming**, financial inclusion is a scaling condition. Many livelihood categories may show opportunity in only a minority of villages, but where opportunity exists, credit access is essential for enterprise growth. With

Energy Access, financial inclusion can support productive electricity use. Villages with good electricity but low finance may be unable to invest in small machinery, processing units, irrigation pumps, cold storage, or rural enterprises.

6.5 Energy Access

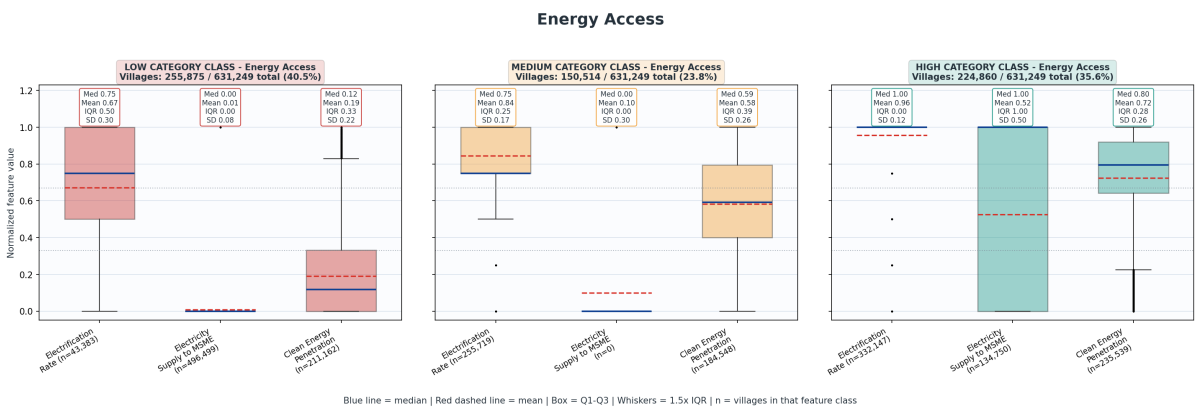

Figure 8: Energy Access

The **Energy Access** category comprises of: **Electrification Rate**, **Electricity Supply to MSME**, and **Clean Energy Penetration**. The methodology calculates Electrification Rate from the reported hours of domestic electricity availability, converting categories into scores: **1–4 hours = 0.25, 4–8 hours = 0.50, 8–12 hours = 0.75, more than 12 hours = 1.00, and no electricity = 0.00**. It captures two different forms of energy access. The first is **domestic electricity availability**, which relates to household welfare, study time, lighting, digital access, appliance use, and general quality of life. The second is **enterprise electricity access**, represented by electricity supply to MSME units. The third is **clean household fuel access**, represented by households using clean energy.

Energy Access has **255,875 villages in Low, 150,514 in Medium, and 224,860 in High**, equivalent to **40.5% Low, 23.8% Medium, and 35.6% High**. The feature-level pattern is very revealing. Electrification Rate is comparatively strong: **332,147 villages are High**, 255,719 are Medium, and only 43,383 are Low. This suggests that domestic electricity availability has reached a large number of villages at moderate to high levels. Clean Energy Penetration is also relatively distributed, with 211,162 Low, 184,548 Medium, and 235,539 High villages. The major weakness is **Electricity Supply to MSME**, where **496,499 villages are Low** and only 134,750 are High. This shows many villages may have domestic electricity access, but not necessarily electricity systems that support rural production, processing, micro-enterprise, or industrial activity. This category therefore may also serve as an indicator for which villages are ready for economic development as electricity supply is as essential to enterprise as any other capital resource.

LOW Energy Access villages

Low Energy Access villages are likely to face one or more of the following constraints: limited domestic electricity hours, low clean-energy household penetration, or absence of electricity supply to MSME units. These villages face both welfare and productive disadvantages.

From a household perspective, low energy access can affect education, digital connectivity, women's workload, indoor air quality, and household health. From a livelihood perspective, it limits small-scale processing, storage, machine use, irrigation pumping, repair services, digital service centres, and cottage or micro-enterprise activity. These villages struggle to move into higher productivity activities such as pump irrigation, post-harvest processing, cold storage, dairy chilling, small machinery, or local agri-enterprise. Low Energy Access becomes especially concerning when combined with low **Housing Quality** and low **Water and Sanitation**. Such villages may have poor household living conditions across multiple dimensions: weaker houses, low clean energy, poor sanitation, drainage gaps, and limited piped water. These should be treated as living-condition deprivation clusters.

MEDIUM Cluster

Medium Energy Access villages may have moderate domestic electricity but incomplete clean fuel coverage or limited productive electricity. Access to clean energy is also related to Ujjwala coverage under Housing Quality and health-related categories. A modest intervention—improving transformer capacity, feeder reliability, MSME electricity provision, LPG access, or clean-energy coverage—may push them into a higher development pathway.

HIGH Energy Access villages

High Energy Access villages have stronger domestic electricity and/or clean energy penetration, and some may also have electricity supply to MSME units. These villages are better positioned for higher-order rural development interventions. However, it does not automatically mean productive transformation. If a High Energy village also has low **Financial Inclusion**, low **Institutionalization**, and low **Agricultural Markets**, then electricity exists but the enabling conditions for enterprise may be missing. In such cases, the priority should not be more energy infrastructure alone; it should be productive use of energy. High Energy Access villages with good roads, strong institutionalization, and higher agricultural activity should be considered for rural enterprise clusters, cold-storage, food processing, dairy chilling, agri-service centres, digital service points, and cottage industry scaling.

6.6 Road Connectivity

The **Road Connectivity** category is built from four features: **All-Weather Road Connection**, **Internal Pucca Road Quality**, **Public Transport Availability**, and **Railway Station Availability**. All-weather road connection and railway station availability are binary features. Internal pucca road quality is scored as **fully covered = 1.00, partially covered = 0.50, and not covered = 0.00**. Public transport availability is derived from the availability of public transport, such as 1: Bus, 2: Van, 3: Auto, 4: None, resulting in binary presence flag. The final Road Connectivity category is the mean of the cluster scores of these four features. It separates **village connection to the outside network**, **internal road quality**, **public transport access**, and **rail connectivity**. This is useful because a village may be connected by all-weather road but still have poor internal roads, weak public transport, or no railway access.

It has **90,511 villages in Low, 275,772 in Medium, and 264,966 in High**, equivalent to **14.3% Low, 43.7% Medium, and 42.0% High**. Feature-level observation shows that All-Weather Road Connection is strong: **431,486 villages are High** and 199,763 are Low. Public Transport Availability is also strong, with **434,518 High** and 196,731 Low. Internal Pucca Road Quality is more mixed: 204,096 Low, 306,365 Medium, and 120,788 High. Railway Station Availability is naturally rare, with 613,871 Low and only 17,378 High. The low prevalence of railway stations should not be interpreted as a universal development failure. Many villages cannot reasonably be expected to have railway access. The more operationally useful features are all-weather road connection, internal pucca roads, and public transport.

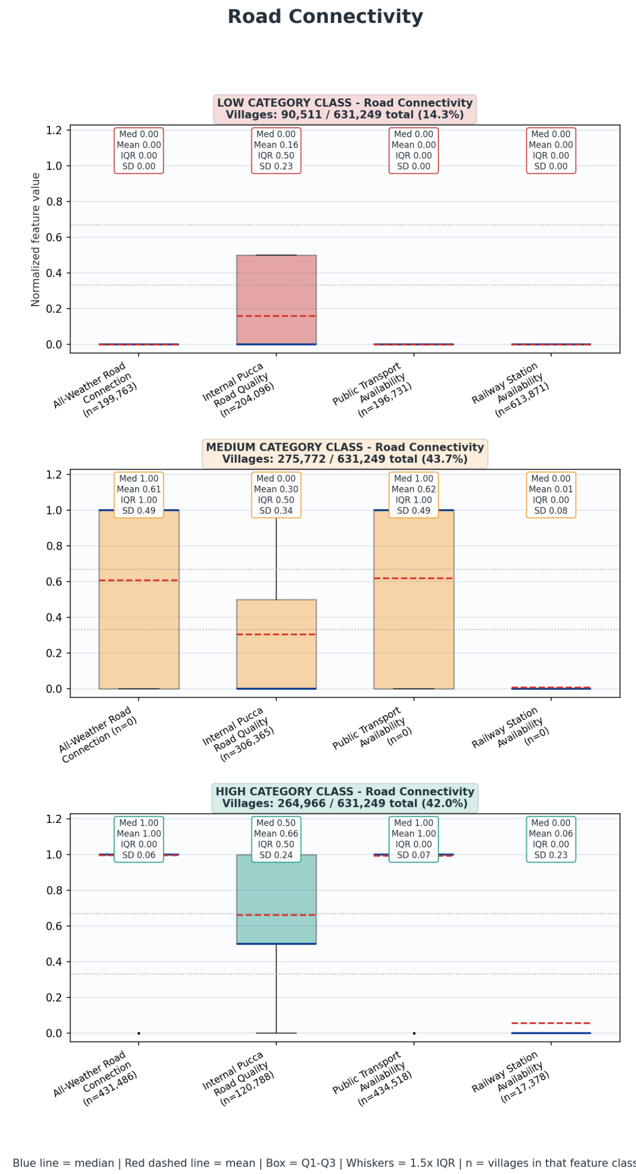

Figure 9: Road Connectivity

LOW Road Connectivity

Low Road Connectivity villages are likely to be physically isolated or poorly connected. They may lack all-weather roads, have weak internal pucca roads, limited public transport, or no rail proximity. Such villages face high transaction costs across nearly every development domain. Low road connectivity affects access to schools, health services, markets, banks, administrative offices, and emergency services. It can reduce teacher attendance, health outreach reliability, commodity movement, and ability to reach welfare or financial service points. From a livelihood perspective, Low Road Connectivity is especially damaging when combined with high **Farm Employment**, high **Agriculture Land Cultivation**, or livelihood potential in **Fisheries**, **Forest Resources**, **Livestock**, or **Cottage Industry**. Without roads and transport, producers cannot easily access inputs or sell outputs. Perishable goods such as milk, fish, vegetables, and forest produce are particularly affected. Low Road Connectivity combined with low **Maternal and Child Health** should also be treated as a service-risk zone.

MEDIUM Road Connectivity Cluster

Medium Road Connectivity is the largest group. These villages may have an external road connection but weaker internal roads, or they may have some transport availability but uneven road quality. This category is crucial because many villages are not completely isolated but are not fully mobility-ready either. Medium villages should be prioritized for internal road upgrading, public transport regularity, road maintenance, and last-mile connection to key facilities such as markets, health centres, schools, banking points, storage centres, and Panchayat facilities.

HIGH Road Connectivity

High Road Connectivity villages have strong mobility potential. They are well suited to serve as service hubs, market access nodes, or logistical centres. But high road connectivity does not automatically translate into development. If a village has high roads but low **Agricultural Markets**, low **Financial Inclusion**, low **Energy Access**, or weak **Institutionalization**, then roads are under-leveraged. High Road Connectivity villages should therefore be used strategically. They can host aggregation centres, storage units, rural haats, financial service camps, producer-group meetings, training centres, veterinary outreach, and health-service delivery points for nearby less-connected villages.

Cross Category Connections

Road Connectivity has one of the strongest multiplier effects across the framework. It is the metric that lets things move. Since large part of human economy is largely exchange of goods, services, everything that requires movement requires roads. With **Agricultural Markets**, road connectivity determines whether farmers can access mandis, haats, storage, and buyers. The analysis shows that Road Connectivity is much stronger than Agricultural Markets. This indicates that many villages may have roads but still lack market systems. In such places, the next priority is not merely road building; it is market activation. Similarly, we can see how same **Financial Inclusion** can have different outcomes, based on availability of roads. With **Social Protection**, roads and transport affect access to Fair Price Shops, pension withdrawal points, administrative offices, and grievance centres. With **Water and Sanitation**, roads support infrastructure construction, pipe laying, drainage work, solid-waste transport, and maintenance. Poor internal roads can make WASH asset management difficult. With **Livestock, Fisheries, Forest Resources, Cottage Industry, and Agriculture**, roads shape the commercial viability of livelihoods. Strong production potential without transport access can trap villages in low-value local markets.

6.7 Housing Quality

The **Housing Quality** category captures household-level living conditions through four dimensions: structural housing condition, access to public housing support, the extent to which PMAY housing demand has been met, and clean cooking-fuel support through Ujjwala. It is therefore not only a “house construction” category. It is better understood as a composite measure of **domestic security, household dignity, scheme reach, and living-environment**. The category is built from four features: **Pucca Housing Rate** (calculated inversely from the share of households with kuccha wall and kuccha roof), **Housing Scheme Coverage** (calculated as households receiving PMAY houses plus households benefiting from state housing schemes divided by total households), **PMAY Demand Met** (calculated as PMAY houses received divided by the sum of PMAY houses received and households in the PMAY permanent wait list), and **Ujjwala Coverage** (measures households availing PMUY benefits divided by total households). The final cluster distribution is: **Low: 225,541 vil-**

lages (35.7%), Medium: 286,500 villages (45.4%), High: 119,208 villages (18.9%). This distribution shows that Housing Quality is neither uniformly weak nor broadly saturated. The largest share of villages lies in the Medium cluster, suggesting that many villages are in a transitional condition: some progress has been made in reducing kuccha housing, expanding housing benefits, or improving clean cooking access, but the full household-living-condition package remains incomplete. The relatively small High cluster, at 18.9%, indicates that strong housing quality is still limited. At the same time, the Low share of 35.7% remains large enough to mark housing as a major rural development concern.

Feature-level distributions show a clear internal pattern. Pucca Housing Rate is the strongest component, with 363,396 villages in High, 166,587 in Medium, and 101,266 in Low. This suggests that the burden of households with both kuccha wall and kuccha roof is relatively lower in many villages. The weakest feature is Housing Scheme Coverage, with 470,271 villages in Low, 126,851 in Medium, and only 34,127 in High. This should not automatically be read as a failure, because not all households require housing assistance. The large Low and Medium counts in PMAY Demand Met show that in many villages, recorded demand remains only partially addressed. These are the villages where housing need is administratively visible but not yet fully resolved. Including Ujjwala in this category is about the recognition that the quality of a dwelling is not just about walls and roof, but also about the environment within it, but results (Ujjwala Coverage LOW cluster village share: 52%) show that this scheme is yet to reach many people, whose homes keep filling up with fatal breathing air.

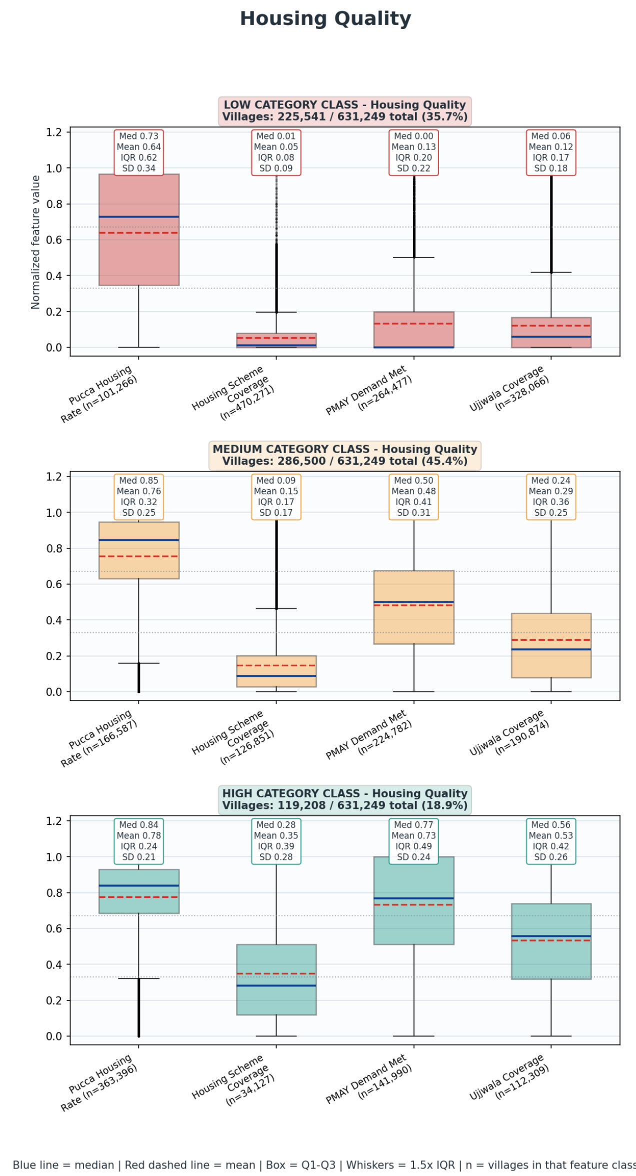

Figure 10: Housing Quality

LOW Housing Quality villages

The **225,541 Low Housing Quality villages** in this cluster comprise over third of all Indian villages. Housing is not merely just another private asset; It shapes health, dignity, safety, education, food, gendered labour, exposure to climate shocks, exposure to WASH contamination, exposure to breathing air, and household resilience. Kuccha housing can increase vulnerability to rain, heat, cold, flooding, and structural insecurity.

Low clean cooking access can increase exposure to smoke and contribute to respiratory risks, especially for women, children, older persons, and those spending more time indoors, and those who cook. Unmet housing demand can also signal that the village contains households whose deprivation is already recognized administratively but remains unresolved. Low Housing Quality villages should also be seen in conjunction with **Water and Sanitation**, **Energy Access**, **Social Protection**, **Financial Inclusion**, and **Maternal and Child Health**. A village with Low Housing Quality and Low WASH is a severe household-environment deprivation cluster. A village with Low Housing Quality and Low Energy Access may face both poor structural conditions and weak household energy transition. A village with Low Housing Quality and Low Social Protection may indicate vulnerable households not sufficiently covered by welfare systems. A village with Low Housing Quality and weak Financial Inclusion may face barriers in documentation, account-linked benefit receipt, DBT, or scheme processing.

MEDIUM Housing Quality

They may have a moderate pucca housing profile but weak scheme coverage. They may have some PMAY progress but continued wait-list pressure. They may have partial Ujjwala coverage but not saturation. The specific intervention depends on which feature is weak. If pucca housing is the weak feature, the village still has a structural housing problem. If Housing Scheme Coverage is weak, the issue may be scheme reach or limited eligibility relative to total households. If PMAY Demand Met is weak, recorded demand requires targeted resolution. If Ujjwala Coverage is weak, the village needs clean-fuel benefit expansion and usage support. The Medium group is also where intervention can be especially effective. In these villages, targeted improvements in PMAY completion, state housing support, Ujjwala coverage, sanitation, piped water, and clean energy can move them into a stronger household-welfare position.

HIGH Housing Quality village cluster

The **119,208 High Housing Quality villages** represent comparatively better household living conditions. These villages likely have lower kuccha housing indicators, better PMAY demand resolution, stronger Ujjwala coverage, or a stronger combination of these features. However, High Housing Quality should not be interpreted as the completion of rural living-condition development. Housing quality depends on the interaction of the dwelling with water, sanitation, electricity, drainage, clean fuel, roads, and livelihood security. A village may score High in Housing Quality while still lacking piped water, drainage, waste disposal, stable electricity, or nearby markets. In such villages, the development agenda can shift from basic housing provision toward household service integration: ensuring that houses are connected to drinking water, sanitation, drainage, electricity, clean energy, and social protection. If High Housing Quality overlaps with strong WASH, Energy Access, and Social Protection, the village may have a strong household welfare base. High villages can also be studied as positive cases. They may reveal administrative patterns associated with better PMAY demand resolution, better uptake of state housing schemes, or stronger Ujjwala coverage. Such villages may serve as models for implementation practices, documentation systems, Panchayat facilitation, or convergence between housing and welfare departments.

Cross-category connection

Housing Quality is **deeply connected to several other categories because the household is the site where many development outcomes are actually experi-**

enced. The strongest relationship is with **Water and Sanitation**. A pucca house without safe water, sanitation, drainage, and waste systems does not provide a complete living environment. A household built by nearby village workers around a chemical plant may result in the children of the workers to play in a 'pond', where their bodies constantly encounter chemical wastages in their day-to-day play. Poor housing, smoke exposure, dampness, overcrowding, and weak sanitation can undermine maternal nutrition, child health, newborn health, and early childhood development, which connects us to **Maternal and Child Health** as well. The category also interacts strongly with **Energy Access**. Ujjwala Coverage directly captures clean cooking support, while Energy Access captures broader electricity and clean-energy penetration. Low Housing Quality generally comes in package with more power cuts. Housing and clean cooking areas should therefore be examined where child underweight, anaemia, stunting, or newborn risk remains high.

The connection with **Social Protection** is equally important. Almost whole state based social security service delivery passes through a mandatory step: **identification**, which even in this country with such high prevalence of housing deprivation, requires everyone to have a verifiable place of stay, in order to get enrolled or while availing any services. Thus housing availability eases up access to social protection, **Financial Inclusion**, and other multipliers. Housing deprivation often overlaps with poverty, old age, disability, widowhood, food insecurity, and other vulnerabilities. If a village has Low Housing Quality and Low Social Protection, households may be missing both physical security and welfare support. Such villages should be prioritized for beneficiary verification, PMAY wait-list review, BPL/NFSA alignment, pension saturation, and household deprivation mapping. **Civic Infrastructure** is also influenced because housing schemes depend on local facilitation: beneficiary identification, public disclosure, wait-list verification, grievance handling, and coordination between households and implementing agencies. Villages with Low Housing Quality and weak Civic Infrastructure may face administrative bottlenecks in scheme delivery.

6.8 Maternal and Child Health

The **Maternal and Child Health** category is one of the most substantively rich categories in the village development framework because it moves beyond a narrow reading of health infrastructure and attempts to represent a care continuum. The category is constructed from five composite features: **AWC Infrastructure and Enrollment Coverage**, **Maternal Health and Care Access**, **Child Nutrition and Development**, **Newborn Health Outcomes**, and **Health Schemes Utilization**. This structure is analytically useful because each feature corresponds to a distinct layer of the maternal-child system. The first layer captures whether the institutional front-end exists and whether children are registered within it. The second captures maternal service contact and risk reduction. The third captures child nutrition, immunization, anaemia, and stunting-related outcomes. The fourth captures newborn health through underweight-at-birth patterns. The fifth captures whether households and eligible beneficiaries are actually connected to PMMVY, PMJAY, and health insurance coverage. The final category classification is generated from these feature-class scores using MiniBatchKMeans, with the distribution showing **103,980 villages in Low (16.5%)**, **213,177 in Medium (33.8%)**, and **314,092 in High (49.8%)**.

At first glance, Maternal and Child Health appears relatively strong compared with many other development categories. Nearly half of all villages fall into the High cluster, and the Low cluster is smaller than in categories such as Financial Inclusion, Housing Quality, Water and Sanitation, Agriculture Support Services, and Agricultural Markets. This suggests that the public care architecture for mothers and young children has a wider reach than many other rural development systems. However, the internal structure of the category prevents a simplistic interpretation. The feature-level results show that the maternal-child system is uneven across its different stages. AWC Infrastructure and Enrollment Coverage is relatively strong, with only 76,223 villages in Low and 234,395 in High, but a very large 320,631 villages in Medium. This indicates that Anganwadi and child enrollment systems are present in many villages but are not uniformly complete. **The most serious internal weakness is Child Nutrition and Development**, where only 57,813 villages are in the High cluster while 208,219 are Low and 365,217 are Medium.

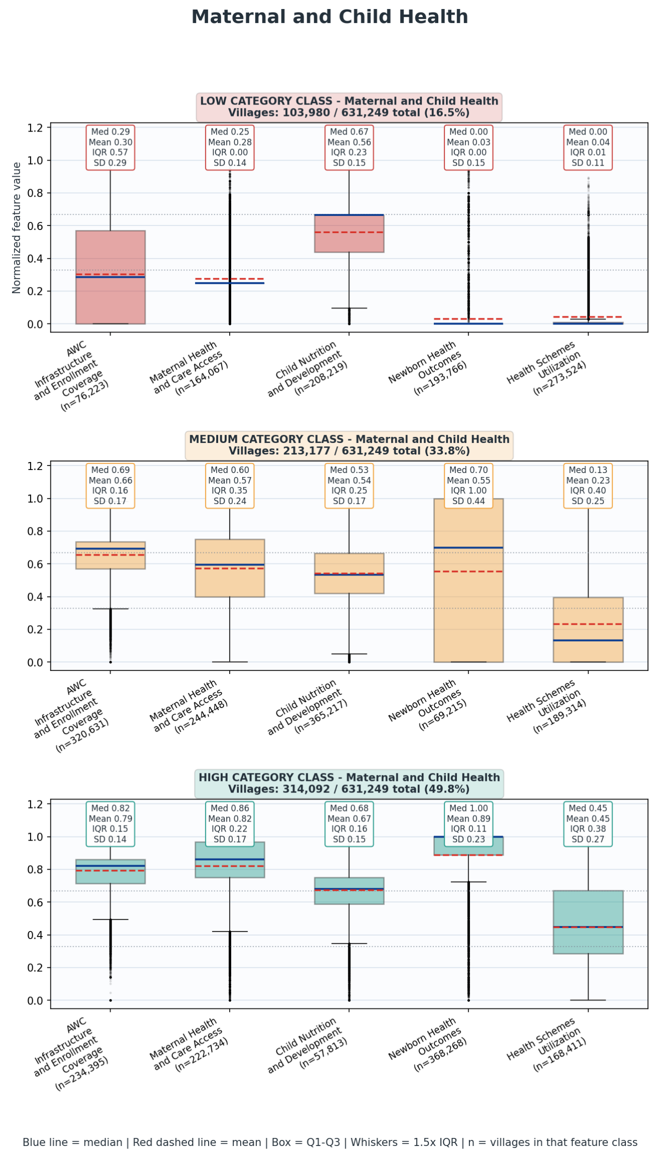

Figure 11: Maternal and Child Health

This tells us that service presence does not automatically translate into strong child nutrition and development outcomes. Newborn Health Outcomes is comparatively stronger, with 368,268 villages in High, but still has 193,766 villages in Low, which should not be ignored. Finally, Health Schemes Utilization is a major bottleneck: 273,524 villages are Low, compared with 168,411 High, indicating that health-related entitlement systems are far from being realized.

The category reveals a familiar development pattern: institutional presence is wider than outcome achievement. Villages may have Anganwadi centres, early childhood education, and registered children, yet still show weak child nutrition, anemia, underweight, stunting-related burdens, or low utilization of PMMVY and PMJAY. This is precisely why the category is valuable. It separates the **presence of the care system** from the **performance of the care system**.

LOW Maternal and Child Health villages

The **103,980 Low villages** should be treated as high-priority maternal-child vulnerability clusters. These villages likely have multiple breaks in the care continuum. Some may lack adequate Anganwadi infrastructure or enrollment. Others may have weak pregnancy and lactation services, high maternal anemia, low institutional delivery, poor child nutrition, weak newborn outcomes, or poor PMMVY/PMJAY utilization. The important point is that a Low village is unlikely to be solved by a single-sector intervention. If the maternal-child system is weak, the underlying causes may include poor WASH, weak housing, low financial inclusion, poor road access, weak civic infrastructure, poor food security, low institutionalization, or poverty. A village with Low MCH and Low WASH, for example, needs a health-nutrition-water-sanitation package. A village with Low MCH and Low Road Connectivity needs referral and transport planning. A village with Low MCH and Low Financial Inclusion needs PMMVY/PMJAY facilitation and banking support. A village with Low MCH and Low Institutionalization may need SHG-based mobilisation, maternal counselling, nutrition campaigns, and beneficiary tracking.

MEDIUM Maternal and Child Health clusters

These villages likely have some functioning maternal-child systems but incomplete outcome translation. The large Medium share suggests that many villages have moved beyond total absence of care infrastructure but remain far from comprehensive care. Medium villages should be treated as care-continuum completion candidates. A village may have good AWC infrastructure but weak child nutrition. Another may have maternal services but weak scheme utilization. Another may have acceptable newborn outcomes but poor child anemia or underweight performance. The policy response must therefore be diagnostic rather than generic. The most useful planning approach for Medium villages is to identify the weakest feature among the five composites.

High Maternal and Child Health villages

The **314,092 High villages** represent comparatively stronger maternal-child systems. They likely have better coverage across infrastructure, maternal care, newborn outcomes, child services, or scheme utilization. However, the category average must not be allowed to hide internal weaknesses. A village can be High overall while still underperforming in Child Nutrition and Development or Health Schemes Utilization.

High villages should therefore not simply be marked as "complete." They should be treated as consolidation and quality-improvement geographies. The planning focus can shift from basic coverage toward service quality, outcome improvement, and institutionalized monitoring. This includes improving early childhood education quality, reducing residual maternal and child anemia, tracking underweight and stunted children, strengthening PMMVY/PMJAY saturation, ensuring complete ICDS-CAS records, and using Panchayat and SHG networks for sustained nutrition behaviour change.

High villages are also useful as potential convergence hubs. If they also have strong Civic Infrastructure, Institutionalization, Road Connectivity, and Financial Inclusion, they may support surrounding Low and Medium villages through training, service camps, nutrition demonstrations, SHG federation support, or health-scheme facilitation.

6.9 Water and Sanitation

The **Water and Sanitation** category is built from five features: **Piped Water Coverage**, **Sanitation Coverage**, **Drainage Quality**, **Community Waste Disposal**, and **Biogas**

or Waste Recycling. Piped Water Coverage is calculated from availability of piped tap water and the household piped-water connection ratio. Sanitation Coverage is based on households not having sanitary latrines divided by total households. Drainage Quality is scored as **closed drainage = 1.00, covered open drainage = 0.75, uncovered open drainage = 0.50, kuchha drainage = 0.25, and no drainage = 0.00**. Community Waste Disposal and Biogas/Waste Recycling are binary features.

This category includes household water access, household sanitation, community drainage, solid-waste disposal, and recycling or biogas systems. It therefore moves beyond toilet construction and captures the wider village sanitation ecosystem. Water and Sanitation has 41.7% Low, 38.4% Medium, and only 19.9% High. Feature-level data shows a very uneven pattern. Sanitation Coverage is strong: the original construction placed 517,603 villages in its internal High band and only 41,849 in Low. This suggests that household sanitary latrine coverage is much better than the broader WASH system. However, Piped Water Coverage is weak, with 305,888 Low, 215,693 Medium, and only 109,668 High villages. Community Waste Disposal is also weak, with 533,797 Low and 97,452 High. Biogas or Waste Recycling is extremely weak, with 593,251 Low and only 37,998 High. Drainage Quality is more mixed: 194,017 Low, 163,940 Medium, and 273,292 High. **This is one of the most important diagnostic findings: the Water and Sanitation category is being pulled down not by sanitation alone, but by piped water, waste disposal, recycling/biogas systems, and drainage gaps.** In other words, toilet coverage has progressed faster than the broader village WASH infrastructure.

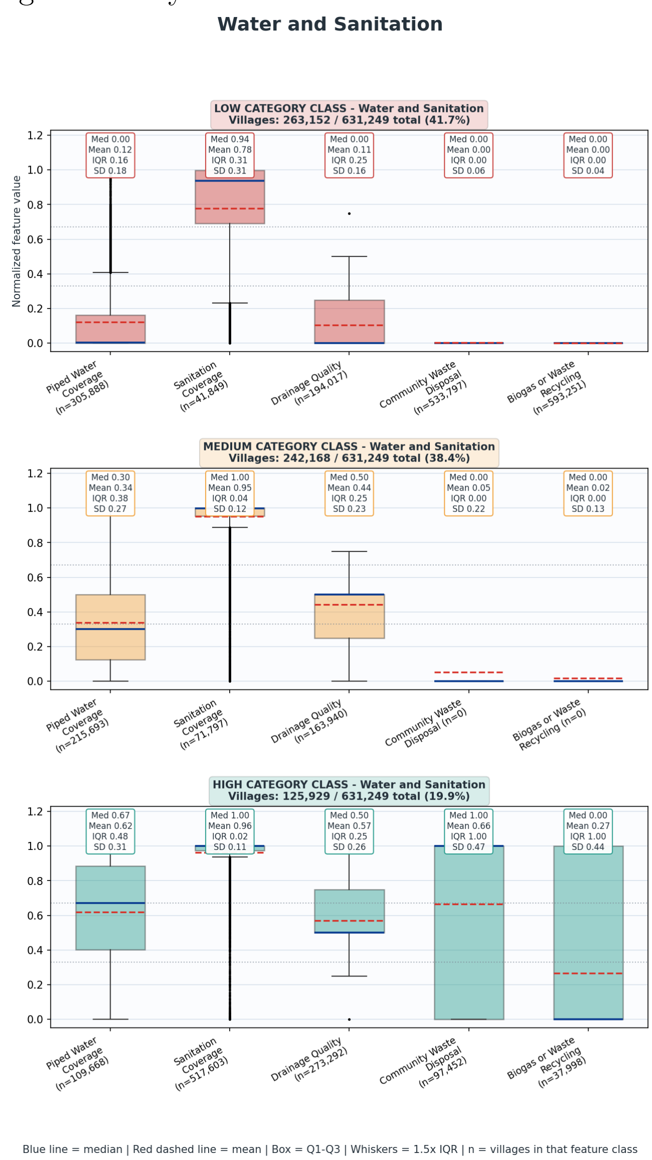

Figure 12: Water and Sanitation

LOW Water and Sanitation villages

Low Water and Sanitation villages are likely to face weak piped water access, poor drainage, weak community waste disposal, absence of biogas or waste recycling, or a combination of these. Even where household toilets exist, the village may still lack the supporting systems for safe water and environmental sanitation. Low WASH is a high-priority condition because it directly affects health, dignity, disease burden, women's workload, child nutrition, and local environmental quality. It should be cross-checked with **Maternal and Child Health**, especially where child underweight, maternal anemia, or low birth weight indicators are weak. Health service delivery alone cannot fully compensate for poor water and sanitation conditions.

MEDIUM Water and Sanitation cluster

Medium Water and Sanitation villages are likely to have partial WASH systems. They may have strong sanitation but weak piped water, or reasonable drainage but poor waste disposal. This is the most common transition group and should be handled through component-specific diagnosis. For Medium villages, the key is not simply "more sanitation." The main gaps are likely to be piped water coverage, drainage quality, solid-liquid waste management, community disposal, and recycling systems. Medium villages can often be upgraded through focused investments in water connections, drainage improvement, and waste infrastructure.

HIGH Water and Sanitation villages

High Water and Sanitation villages have stronger WASH systems, but even these should be examined feature-wise. A village may be High because it performs well on sanitation and drainage, while still having gaps in piped water or recycling. The category should be used as a starting point, not a substitute for detailed WASH planning. High WASH villages can be used as models for integrated rural sanitation: household latrines, piped water, drainage, waste disposal, and recycling. If they also have high Civic Infrastructure, they may be good demonstration sites for Panchayat-led maintenance systems. High WASH also strengthens outcomes in health, housing, and livelihoods. Clean water and drainage reduce disease burden; waste systems improve environmental conditions; and reliable water supports schools, Anganwadis, community facilities, and small enterprises.

6.10 Livelihoods Cottage and Traditional Industry

The **Livelihoods Cottage and Traditional Industry** category is built from four features: 1. **Cottage Units Available**, 2. **Cottage Industry Participation**, 3. **Handloom**, and 4. **Handicrafts**. Features 1, 2, 4 as listed above are binaries and Cottage Industry Participation (2) is calculated as households engaged in cottage or small-scale units divided by total households. This category is highly concentrated. The final category distribution is **587,952 villages in Low, 0 in Medium, and 43,297 in High**, or **93.1% Low and 6.9% High**. The feature-level pattern confirms that cottage and traditional industry presence is rare across villages.

The most important note about this category is that cottage and traditional industry is not a universal village feature. It appears in specific pockets. Therefore, the High cluster is extremely important: it identifies villages where a livelihood base already exists and can potentially be strengthened through credit, design, training, raw material supply, producer collectives, market access, storage, branding, and digital commerce.

LOW village clusters

Low villages in this category should not automatically be treated as deficient. In many villages, cottage industry, handloom, and handicrafts may not be historically present, economically viable, or locally demanded. A generic push to create cottage industries everywhere may result in weak adoption, poor market fit, and unsustainable subsidy-led activity.

However, Low cottage-industry villages become important when they have other enabling strengths. For example, a Low village with high **Road Connectivity**, good **Energy Access**, strong **Financial Inclusion**, and strong **Institutionalization** may be a candidate for new rural enterprise development, even if traditional industry is not currently present. Conversely, a Low village with weak roads, weak finance, weak institutions, and weak market access should not be pushed immediately into enterprise promotion without first building the enabling system. Low villages should therefore be divided into two sub-groups: villages where cottage industry is not relevant, and villages where enterprise potential exists but has not yet been activated.

HIGH Clusters

High villages are the core opportunity set. These villages already have cottage units, household participation, handloom, handicrafts, or a combination of these. The development question is no longer whether livelihood potential exists; it is mostly about whether the ecosystem around that potential is strong enough. High cottage-industry villages should be cross-checked with **Financial Inclusion**, **Institutionalization**, **Energy Access**, **Road Connectivity**, and **Agricultural Markets**. A High cottage-industry village with low financial inclusion may have artisans and workers but lack working capital. A High cluster village with low institutionalization may have individual workers but weak collective bargaining, poor aggregation, and limited access to schemes. A High village with weak roads or markets may struggle to sell outside local demand. A High village with weak energy access may be unable to adopt tools, machines, packaging units, or digital operations.

Therefore, High villages should be treated as **livelihood-cluster strengthening zones**. For High villages, prioritize cluster development: artisan mapping, SHG/producer-group formation, working capital, design support, toolkits, common facility centres, branding, packaging, digital market access, and links with local haats and larger buyers. For **High cottage-industry + low finance/institutionalization villages**, start with SHG-bank linkage, credit support, and collective formation. For **High cottage-industry + high roads/energy villages**, push market expansion, processing, digital commerce, and value addition.

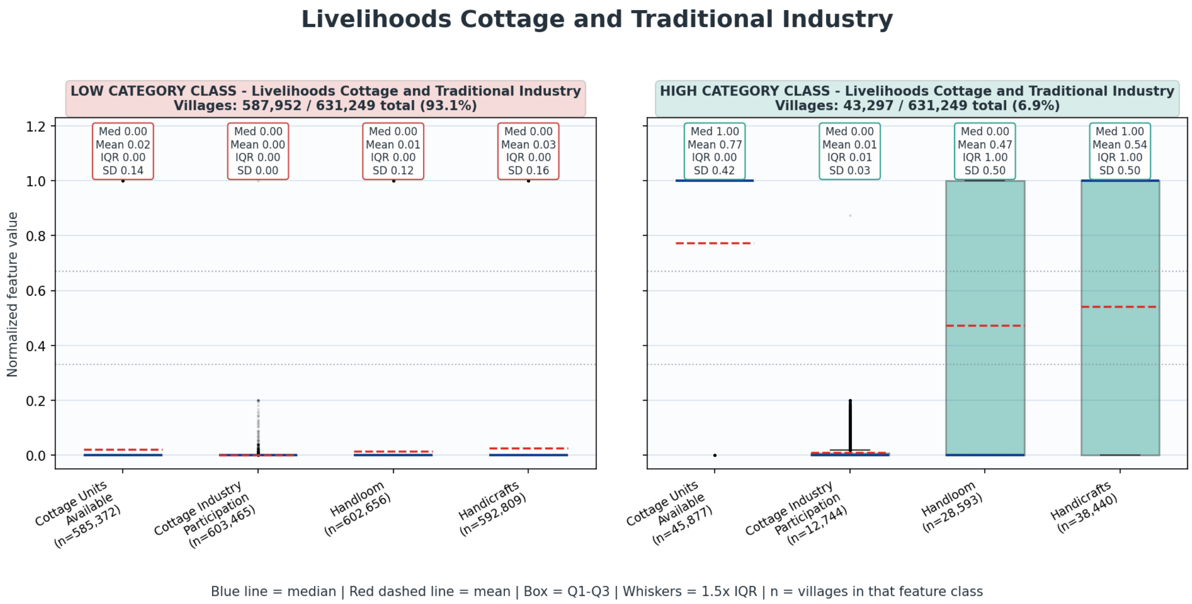

Figure 13: Livelihoods Cottage and Traditional Industry

6.11 Farm Employment

The **Farm Employment** category is built from a single feature: **Farm Employment**, calculated as share of village households engaged in farm activities. This category should not be interpreted as a simple "good" or "bad" score. High farm employment can mean strong agricultural engagement, but it can also mean high dependence on agriculture and limited non-farm diversification. Low farm employment can mean either a non-agricultural economy, landlessness, occupational diversification, migration, or weak local agricultural opportunity. Farm Employment has **144,994 villages in Low, 197,800 in Medium, and 288,455 in High**, equivalent to **23.0% Low, 31.3% Medium, and 45.7% High**. This is one of the more widely distributed livelihood categories, and unlike fisheries, forest livelihoods, or cottage industries, it applies to a much larger share of villages. The relatively high share of High villages confirms the centrality of agriculture in village livelihoods. However, it should be read alongside **Agriculture Land Cultivation**, **Agriculture Irrigation and Watershed**, **Agriculture Support Services**, **Agricultural Markets**, and **Financial Inclusion**. Farm employment alone does not tell us whether agriculture is productive, remunerative, irrigated, market-linked, or resilient.

LOW Farm Employment

Low Farm Employment villages may have fewer households engaged in agriculture. This could indicate occupational diversification, non-farm employment, migration, service-sector work, landlessness, forest/fisheries dependence, or simply limited cultivable land. Therefore, Low is not automatically negative. We instead need to examine what replaces farm employment. If Low Farm Employment is accompanied by strong cottage industry, fisheries, livestock, markets, and financial inclusion, the village may have diversified livelihoods. If Low Farm Employment is accompanied by weak non-farm livelihoods, weak markets, low financial inclusion, and low social protection, it may indicate livelihood vulnerability or out-

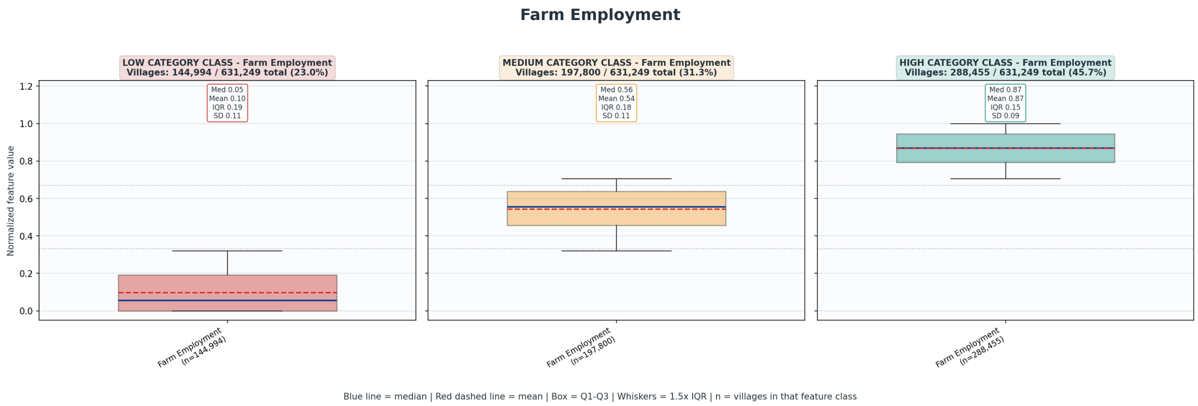

Figure 14: Farm Employment

migration.

Low Farm Employment villages should also be checked against **Housing Quality** and **Social Protection**. If farm employment is low and social protection is weak, households may lack both agricultural income and safety nets.

MEDIUM Farm Employment villages

Medium villages represent mixed livelihood structures. Agriculture is present but not dominant for all households. These villages may be well suited for livelihood diversification strategies: agriculture plus dairy, fisheries, cottage industry, small enterprise, or wage employment. Medium villages should be diagnosed based on their enabling conditions. **If roads, finance, and energy are strong, non-farm enterprise can be promoted. If land cultivation and irrigation are strong, agriculture intensification may be viable.** If common pastures and livestock services exist, livestock can be strengthened. If markets are weak, the first intervention should be aggregation and market access.

HIGH Farm Employment Clusters

High Farm Employment villages are agriculture-dependent villages. They are not automatically high-income villages. Their future depends on whether agriculture is supported by irrigation, watershed development, soil testing, input access, insurance, credit, producer institutions, storage, and markets. The most concerning pattern is **High Farm Employment + Low Agriculture Support Services + Low Agricultural Markets + Low Financial Inclusion**. These villages may have many households dependent on farming but weak support systems. This is a structural vulnerability cluster. The strongest opportunity pattern is **High Farm Employment + High Land Cultivation + Medium/High Institutionalization + Good Roads**. These villages are strong candidates for FPOs, producer groups, aggregation, irrigation improvement, soil testing, crop planning, storage, and market linkage.

6.12 Livelihoods Forest Resources

The category is built from three features: **Community Forest (binary)**, **Minor Forest Production (binary)**, and **Forest Dependence** (as households whose only source of livelihood is minor forest production divided by total households). This makes **Livelihoods**

Forest Resources is a specialized category, with only 8.9% of villages in the High cluster. This should be interpreted as an opportunity and vulnerability marker rather than a universal development score. High villages indicate the presence of community forest, minor forest produce, or household dependence on forest-based livelihoods. These villages need targeted planning around sustainable harvesting, community rights, aggregation, SHG or producer-group formation, storage, processing, credit, and market linkage. Low villages should not automatically be treated as deficient, because forest resources are naturally location-specific.

Feature counts show that Community Forest is High in **96,754 villages**, Minor Forest Production is High in **77,317 villages**, but Forest Dependence is High in only **14,324 villages** and Medium in **9,846**. It is important to note that Forest resources may exist in many more places than households exclusively dependent on minor forest produce. The category clearly identifies a smaller subset of villages where forest-resource presence and livelihood relevance combine.

LOW Forest Resources villages

Low Forest Resources villages should not be interpreted as underdeveloped. Many villages simply may not have forest resources. For these villages, forest-based livelihood planning may be irrelevant or low priority. Low Forest Resources villages should naturally be directed toward other livelihood pathways: agriculture, livestock, fisheries, cottage industry, wage employment, or non-farm enterprise depending on local conditions. In case they are LOW in other livelihood indicators as well, and they have forest resources present, it may require specific planning for the village, based on local situations.

HIGH Forest Resources village clusters

High Forest Resources villages are specialized livelihood-opportunity villages. They may have community forest, minor forest produce, or households dependent on minor forest production. These villages require a different development approach from normal agricultural villages. The main risk is that forest-dependent households may remain in low-value extraction without processing, storage, rights support, aggregation, or market access. If High Forest Resources overlaps with low **Financial Inclusion**, low **Road Connectivity**, low **Institutionalization**, and low **Agricultural Markets**, forest produce may be sold through informal channels at low prices. High Forest villages should therefore be prioritized for minor forest produce value chains, community forest governance, SHG or producer-group formation, storage, primary processing, sustainable harvesting, transport, and buyer linkage.

Interplay with other categories

Forest Resources strongly relates to **Institutionalization**. Collective action is crucial for sustainable harvesting, aggregation, bargaining, and processing. SHGs and producer groups can be especially important for women-led forest produce livelihoods. It also interacts with **Road Connectivity**. Forest produce is often perishable or bulky, and poor connectivity reduces market options. Low-road forest villages may require to take a different approach: local processing, collection centres, or mobile procurement systems. **Financial Inclusion** is essential for working capital and reducing dependence on informal traders. **Agricultural Markets** matters because local haats, storage, and market access can be adapted for forest produce.

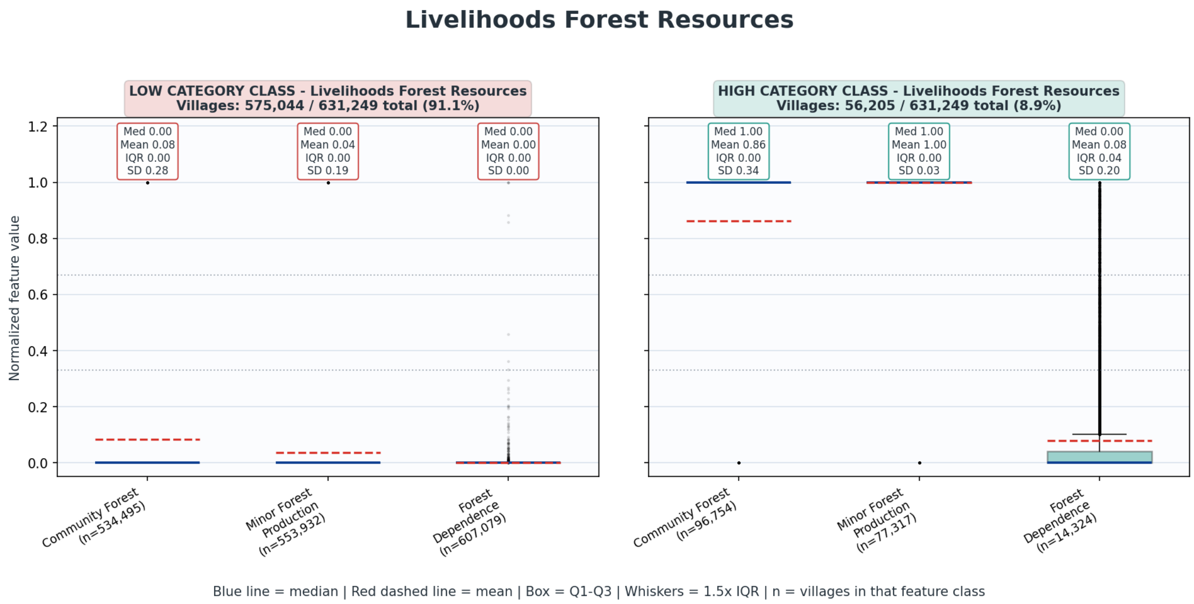

Figure 15: Livelihoods Forest Resources

6.13 Livelihoods Common Resources

The **Livelihoods Common Resources** category is built from one feature: **Common Pastures**. This is a binary feature based on whether common pastures are available as per revenue records. The final category is the cluster score of this feature. Livelihoods Common Resources is a simple but important category based on common pasture availability. With 22.7% of villages in the High cluster, it identifies a meaningful set of villages where pasture-based livestock development may be feasible. This category should be read with Livestock and Veterinary, Financial Inclusion, Road Connectivity, Water and Sanitation, and Watershed Development. High common-pasture villages with weak veterinary services or milk routes represent an important missed opportunity for livestock-based livelihoods.

LOW Common Resources

Low villages may lack common pastures. This does not necessarily mean they are underdeveloped, because pasture availability depends on geography, land records, settlement patterns, and land-use history. However, absence of common pastures matters if the village has strong livestock dependence or poor fodder access. Low Common Resources combined with high **Livestock and Veterinary** may indicate that livestock activity exists but depends on private fodder, purchased feed, crop residues, or external grazing. Low Common Resources combined with low **Livestock and Veterinary** may indicate limited livestock livelihood potential.

High Common Resources villages

High Common Resources villages are important livestock opportunity zones. Common pastures can support dairy, goatery, sheep rearing, fodder systems, and community-managed grazing. But pasture availability alone does not create income. It must be linked with veterinary services, milk routes, livestock development projects, water access, roads, and finance. The most actionable pattern is **High Common Pastures + Low Livestock and**

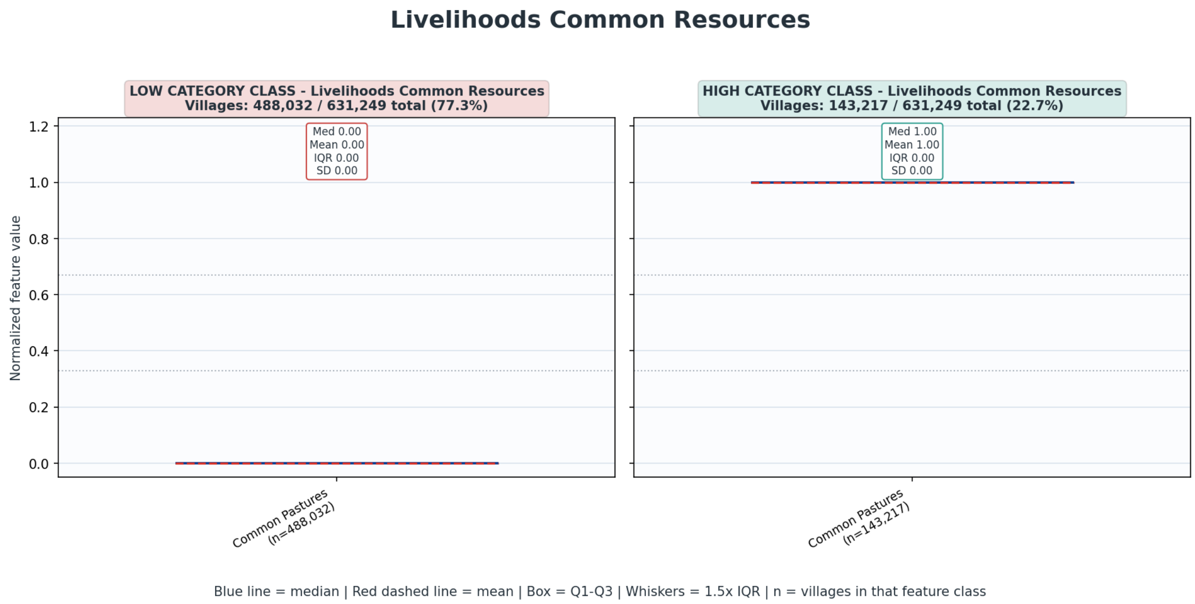

Figure 16: Livelihoods Common Resources

Veterinary. This indicates a resource base without adequate service support. Such villages should be prioritized for veterinary outreach, livestock extension, milk collection routes, fodder development, pasture improvement, and livestock credit. High Common Resources villages can also support climate resilience and land-based ecological planning. Pasture restoration, soil conservation, fodder plantations, and water harvesting can improve both livelihoods and environmental conditions.

6.14 Livelihoods Alternative Farming

The **Livelihoods Alternative Farming** category is built from one composite feature: **Alternative Farming**. It is calculated as a composite binary share of **Bee Farming** and **Sericulture**. Livelihoods Alternative Farming is highly specialized, with only 4.7% of villages in the High cluster. This category should be treated as a niche opportunity indicator, not a universal deficit measure. Villages with beekeeping or sericulture presence should be prioritized for technical extension, input support, producer organization, credit, processing, branding, and market linkage. Expansion should be based on ecological suitability, skill availability, and buyer access rather than uniform promotion.

For beekeeping, support may include hive access, technical training, disease management, linkages with horticulture or forest systems, processing, packaging, branding, and buyer linkage. For sericulture, support may include host-plant development, rearing infrastructure, technical extension, cocoon marketing, and working capital.

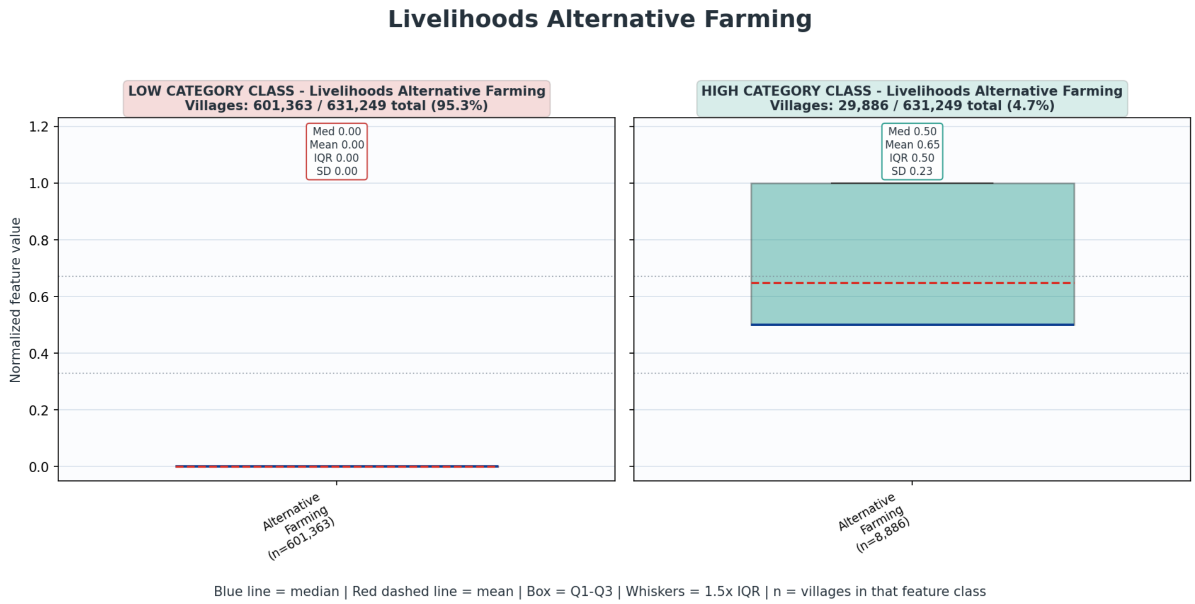

Figure 17: Livelihoods Alternative Farming

6.15 Livelihoods Fisheries

Livelihoods Fisheries is a context-specific opportunity category, with 9.7% of villages in the High cluster and 11.3% in Medium. It identifies villages where fish farming, community ponds, or aquaculture extension facilities are present. Fisheries should not be promoted uniformly across all villages; it should be targeted where water bodies, technical support, and market viability exist. High and Medium fisheries villages should be linked with credit, extension, pond management, cold-chain, road connectivity, markets, and producer organization.

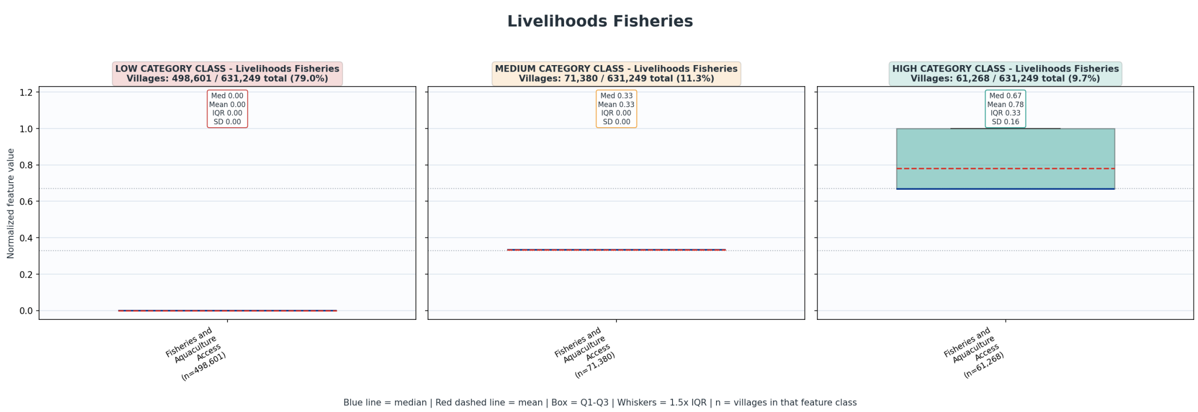

Figure 18: Livelihoods Fisheries

This distribution is context-specific. Fisheries depends on ponds, water bodies, coastal or inland fishery systems, aquaculture suitability, and extension access. The 21.0% of villages in Medium or High are therefore important potential geographies for fishery-based livelihood planning.

LOW Fisheries villages may lack fish farming, community ponds, or aquaculture extension facilities. This is not automatically a development gap. Many villages may not have suitable water bodies or fishery traditions. **However, Low Fisheries villages with significant water bodies not captured in the data may need validation. Low Fisheries combined with high irrigation/watershed assets might indicate untapped potential, especially where ponds or water harvesting structures exist but are not used for fishery livelihoods.**

MEDIUM villages may have one or two components of the fisheries ecosystem but not all. For example, a village may have fish ponds but no aquaculture extension facility, or fish farming but weak community pond use. These villages are ideal candidates for system completion. The key action in Medium villages is to identify the missing element: water body availability, community pond management, extension services, seed/fingerling supply, feed, credit, cold-chain, or market linkage.

HIGH Fisheries villages are strong fishery livelihood opportunity zones. They likely have fish farming, fish community ponds, aquaculture extension, or a combination of these. But fisheries livelihoods are highly sensitive to service and market systems. High Fisheries villages require credit, technical support, disease management, pond management, feed supply, cold-chain, storage, roads, and buyer linkage. If High Fisheries overlaps with low **Road Connectivity** or low **Agricultural Markets**, fish producers may face major losses due to perishability. If it overlaps with low **Financial Inclusion**, farmers may lack working capital for seed, feed, and pond preparation.

6.16 Livestock and Veterinary

The **Livestock and Veterinary** category is built from three features: **Veterinary Services Access**, **Livestock Development Projects**, and **Milk Routes**. Veterinary Services Access is scored as the mean of veterinary hospital availability and livestock extension services. Livestock Development Projects is a composite based on poultry development, goatery development, and piggery development projects. Milk Routes is a binary feature based on availability of milk collection centres, milk routes, or chilling centres. This category captures three domains of livestock ecosystem: service access, project support, and market/logistics linkage.

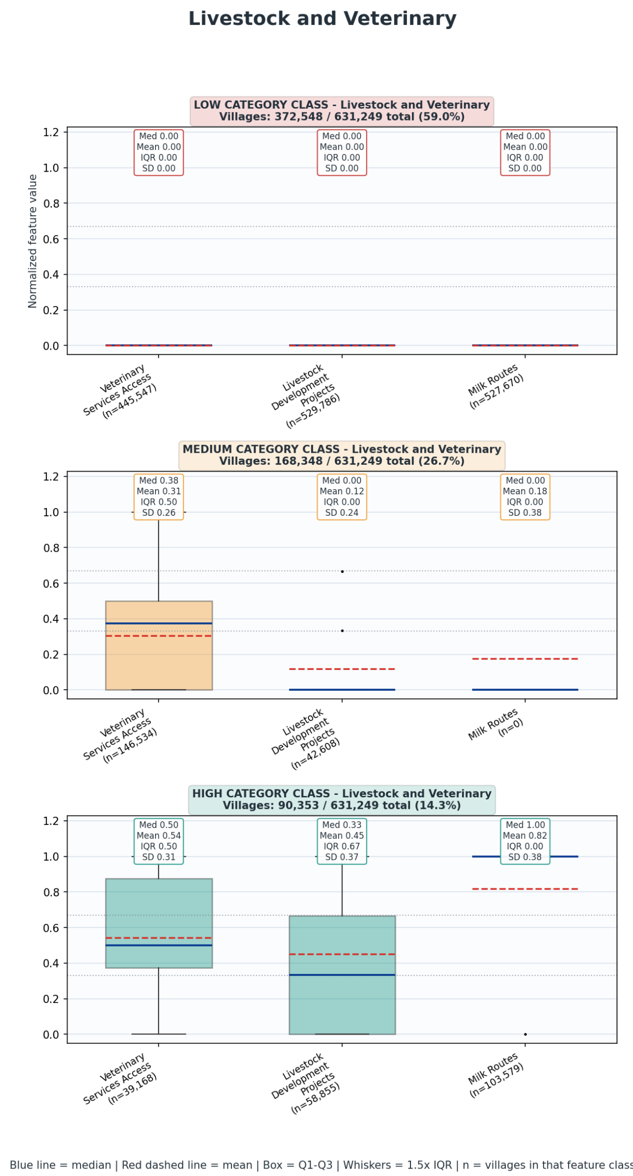

Figure 19: Livestock and Veterinary

Livestock and Veterinary has **372,548 villages in Low, 168,348 in Medium, and 90,353 in High**, equivalent to **59.0% Low, 26.7% Medium, and 14.3% High**. Feature-level metrics show that the weakest part is Veterinary Services Access with only 39,168 High. Livestock Development Projects are also limited: 529,786 Low. Milk Routes are present in more villages than the other two High counts, but still limited: 527,670 Low.

Livestock and Veterinary is a major livelihood-support category, but 59.0% of villages remain in the Low cluster. The weakness is driven by limited veterinary service access, limited livestock development projects, and limited milk-route infrastructure. This cluster matters because livestock can provide income diversification, risk buffering, nutrition, and women-led livelihood opportunities. The category should be analysed with Common Resources, Farm Employment, Financial Inclusion, Road Connectivity, Energy Access, Institutionalization, and Agricultural Markets. Villages with pasture availability but weak livestock services are especially important missed-opportunity zones.

LOW Livestock and Veterinary villages

Low villages are likely to lack veterinary services, livestock extension, development projects, and milk-route infrastructure. This can severely limit livestock productivity, disease control, breed improvement, dairy income, poultry/goatery/piggery expansion, and market participation. Low Livestock and Veterinary becomes especially critical when the village has **Common Pastures**. A village may have grazing resources but weak animal health and market systems. This is a missed livelihood opportunity. Low cluster support also matters in high **Farm Employment** villages. Livestock often acts as a supplementary income source, risk buffer, and nutrition source for farm households. Weak veterinary systems can reduce resilience.

MEDIUM Livestock and Veterinary villages

Medium villages may have partial livestock systems. They might have milk routes but weak veterinary care, or development projects without strong market linkage. These villages are good candidates for system strengthening because some base already exists. The key is to identify the missing service layer. If milk routes exist but veterinary services are low, animal productivity and milk quality may remain weak. If veterinary services exist but milk routes are absent, households may keep livestock but lack reliable market access. If development projects exist but finance is weak, uptake may remain limited.

HIGH Livestock and Veterinary villages

High villages are livestock-support hubs. They may have veterinary access, livestock projects, and milk routes. These villages are suitable for dairy development, goatery, poultry, piggery, fodder planning, breed improvement, animal health camps, and producer-group formation. However, High Livestock and Veterinary should still be checked against **Financial Inclusion**, **Common Resources**, **Road Connectivity**, and **Agricultural Markets**. Livestock livelihoods require credit, feed, veterinary care, transport, chilling, market access, and risk protection. High villages can also support surrounding villages as veterinary outreach or milk-collection nodes, especially if roads are strong.

6.17 Agriculture Land Cultivation

The **Agriculture Land Cultivation** category is built from one feature: **Land Utilization**. This feature uses a composite land-use score based on cultivable area, net sown area, irrigated area, and seasonal sown area. The land utilization score is calculated as the mean of ratios for net sown area over total cultivable area, irrigated area over total cultivable area, and cropping intensity. This single feature is built are all land-improvement variables: total_cultivable_area_in_hac, net_sown_area_in_hac, area_irrigated_in_hac, and net sown area during Kharif, Rabi, and other seasons. This means the category is not only measuring whether agricultural land exists. It is measuring how actively the cultivable land base is being used, whether irrigation contributes to that use, and whether cropping extends beyond a single-season pattern.

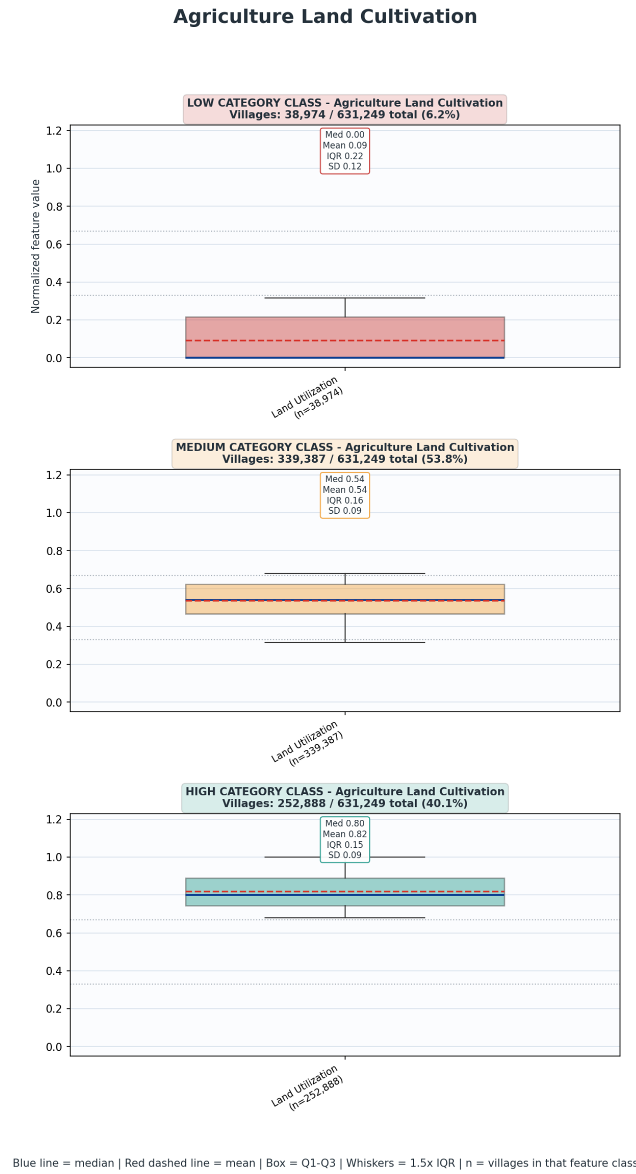

Figure 20: Agriculture Land Cultivation

Agriculture Land Cultivation is one of the strongest agriculture-related categories. It has **38,974 villages in Low, 339,387 in Medium, and 252,888 in High**, equivalent to **6.2% Low, 53.8% Medium, and 40.1% High**. This means that most villages are not land-use deficient in a basic sense. Nearly 94% of villages fall in Medium or High. The central development problem is therefore not simply that land is unused; it is whether cultivated land is supported by irrigation, inputs, risk protection, soil testing, credit, institutions, storage, and markets. In simple words, **the agricultural base is much stronger than the agricultural support ecosystem**.

LOW Agriculture Land Cultivation villages

Low Land Cultivation villages are a small but important group. They may have low cultivable land utilization, limited irrigated area, weak cropping intensity, or land-use constraints. However, Low should not automatically be interpreted as failure. Some villages may be forested, hilly, pastoral, water-scarce, urbanizing, or dependent on non-farm livelihoods rather than cultivation. Low Land Cultivation villages should therefore be profiled carefully. If they also show high **Forest Resources**, **Fisheries**, **Common Resources**, **Cottage Industry**, or **Livestock and Veterinary** potential, then agriculture may not be the main pathway. **If they show low scores across both agriculture and alternative livelihood categories, they may be livelihood-vulnerable and may need wage employment, skill development, social protection, and non-farm enterprise support.**

MEDIUM Agriculture Land Cultivation Villages

The Medium group is the largest. These villages have an active agricultural base but not the strongest land-use intensity. They are the most important scale opportunity for improvement. Medium villages may be limited by irrigation, seasonal cropping, soil quality, input access, farmer risk support, or market incentives. For these villages, the priority is not simply to expand cultivation, but to improve productivity and resilience. This includes irrigation planning, watershed work, cropping-system improvement, soil testing, input availability, insurance, credit, and market access. Medium land-cultivation villages are especially important where **Farm Employment** is high. If many households depend on farming but land utilization is only medium, there may be scope for improving cropping intensity, irrigation reliability, soil management, and farm productivity.

HIGH Agriculture Land Cultivation villages

High Land Cultivation villages have strong agricultural land-use performance. These villages likely have more active use of cultivable land, better cropping intensity, stronger irrigated area contribution, or a combination of these. However, High Land Cultivation should not be read as high agricultural prosperity. A village may cultivate land intensively but still have weak irrigation sustainability, poor support services, low insurance coverage, low soil testing, weak markets, poor storage, low financial inclusion, or weak producer institutions. The most important planning use of High Land Cultivation villages is to identify where returns can be increased through support systems. If High Land Cultivation overlaps with low **Agriculture Support Services** and low **Agricultural Markets**, the village is likely producing without adequate advisory, risk protection, input support, storage, or price realization. These are priority villages for productivity and value-chain interventions.

The main development challenge is not simply bringing land into cultivation, but improving the systems around cultivation. The category should be read with Farm Employment, Irrigation and Watershed, Agriculture Support Services, Agricultural Markets, Financial Inclusion, and Institutionalization. Villages with strong land cultivation but weak support services and market access represent high-priority productivity and value-chain intervention zones.

6.18 Agriculture Irrigation and Watershed

The **Agriculture Irrigation and Watershed** category combines three features: **Irrigation Infrastructure and Watershed Development**, **NREGA NRM Expenditure**, and **Modern Irrigation**. We calculate the irrigation/watershed infrastructure feature as a composite binary share of canal presence, rainwater harvesting systems, and watershed development projects. NREGA NRM Expenditure is calculated as ration of 'expenditure approved under natural resource management in the labour budget' to 'the total approved labour budget'. Modern Irrigation is calculated as 'farmers using drip or sprinkler irrigation' divided by 'total farmers'.

Agriculture Irrigation and Watershed has **263,613 villages in Low, 175,705 in Medium, and 191,931 in High**, equivalent to **41.8% Low, 27.8% Medium, and 30.4% High**. **This contrasts sharply with the strong Agriculture Land Cultivation category.** While only 6.2% of villages are Low in land cultivation, 41.8% are Low in irrigation and watershed. This means many villages are agriculturally active without an adequately strong water-security system.

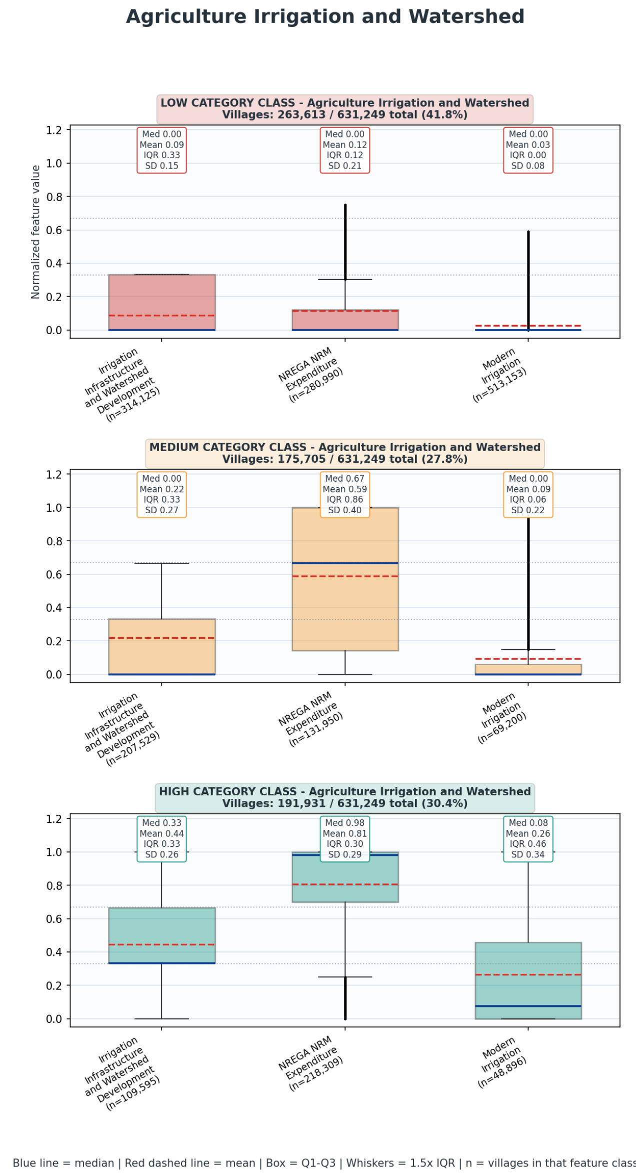

Figure 21: Agriculture Irrigation and Watershed

The feature-level metrics show that Modern Irrigation is the weakest component, with only 48,896 in High. Irrigation Infrastructure and Watershed Development is also limited, with 314,125 Low. NREGA NRM Expenditure is more distributed, with 280,990 Low, 131,950 Medium, and 218,309 High. This suggests that NREGA/NRM spending has reached many villages, but modern irrigation adoption and physical irrigation/watershed infrastructure remain uneven. Spending alone may not be translating into farmer-level water-use efficiency.

LOW Irrigation and Watershed villages

Low Irrigation and Watershed villages may lack canal-linked irrigation, rainwater harvesting systems, watershed development, NREGA NRM investment, and drip/sprinkler adoption. These villages face agricultural risk, especially where farm employment and land cultivation are high. The most concerning pattern is **High Farm Employment + Medium/High Land Cultivation + Low Irrigation and Watershed**. In such villages, many households depend on farming, land is being cultivated, but water security is weak. These villages are exposed to rainfall variability, crop failure, low cropping intensity, and limited diversification.

MEDIUM Irrigation and Watershed villages

Medium villages may have some water infrastructure or NREGA investment but weak modern irrigation adoption. They may also have watershed works but limited farmer-level use of drip or sprinkler systems. These villages are strong candidates for water-productivity improvement. The key question is whether the gap is infrastructure, expenditure allocation, or adoption. If NREGA NRM spending is high but modern irrigation remains low, then farmer technology adoption and extension may be the bottleneck. If modern irrigation is present but watershed and rainwater systems are weak, then recharge and water-source sustainability may be the concern. Medium villages should be approached with a combined asset-and-adoption strategy: watershed treatment, farm ponds, check dams, rainwater harvesting, canal repair, micro-irrigation, farmer training, and crop planning.

HIGH Irrigation and Watershed villages
High Irrigation does not automatically mean higher farmer income. Water availability must be linked to inputs, soil testing, risk support, markets, storage, and producer institutions. High Irrigation villages with low **Agricultural Markets** may have production potential but weak price realization. High Irrigation villages with low **Agriculture Support Services** may have water but not the technical support needed for efficient cropping. These villages should be targeted for value-chain upgrading, not only irrigation maintenance.

6.19 Agriculture Organic Farming

The **Agriculture Organic Farming** category is built from one feature: **Organic Farming**. Agriculture Organic Farming is a specialized adoption category, with 85.7% of villages in the Low cluster and only 5.3% in High. This should not be treated as a universal deficiency, because organic farming requires technical support, farmer confidence, input alternatives, aggregation, certification, and market linkage. The most important use of this category is to identify existing and emerging organic-adoption clusters. High and Medium villages should be supported through soil testing, organic input systems, SHG/FPO aggregation, certification support, branding, and market access. Low villages should be screened for suitability before promotion.

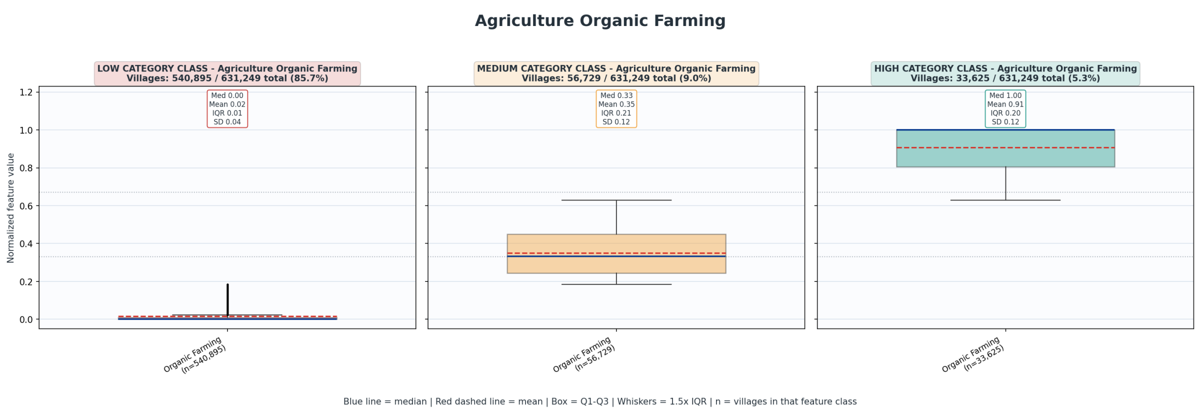

Figure 22: Agriculture Organic Farming

MEDIUM Organic Farming villages represent emerging organic-adoption geographies. These may have some farmers adopting organic practices but not enough to create a strong village-level cluster. The planning priority for Medium villages is consolidation. This includes farmer training, input preparation, composting, bio-input support, soil testing, peer learning, SHG/FPO aggregation, and market testing. Medium villages are suitable for pilot scaling if they also have good roads, markets, institutionalization, and financial access.

HIGH Organic Farming villages are specialized opportunity areas. They have stronger farmer adoption and may be suitable for organic clusters, certification support, branding, aggregation, processing, and market linkage. However, High Organic villages must be checked against **Agricultural Markets** and **Institutionalization**. Organic production without collective aggregation and buyer linkage can fail to produce higher farmer incomes. It must also be checked against **Agriculture Support Services**, because organic farming requires soil knowledge, pest management, input substitution, and farmer advisory. High Organic villages should not only be celebrated as adoption zones; they should be converted into value-chain zones.

6.20 Agriculture Support Services

Agriculture Support Services should be understood as the **capability infrastructure of farming** rather than as a measure of cultivation itself. This cluster captures three linked forms of support: physical access to input and advisory institutions through fertilizer shops, government seed centres, and soil testing centres; formal risk and social-security coverage through PMFBY and PM Kisan Pension Yojana; and adoption of soil-test-based fertilizer application. It asks whether farmers are supported by the technical, institutional, and risk-protection systems required to farm productively and securely. In theoretical terms, this category represents the transition from agriculture as subsistence or inherited practice to agriculture as an informed, institutionally supported, risk-aware, and potentially productivity-enhancing livelihood system.

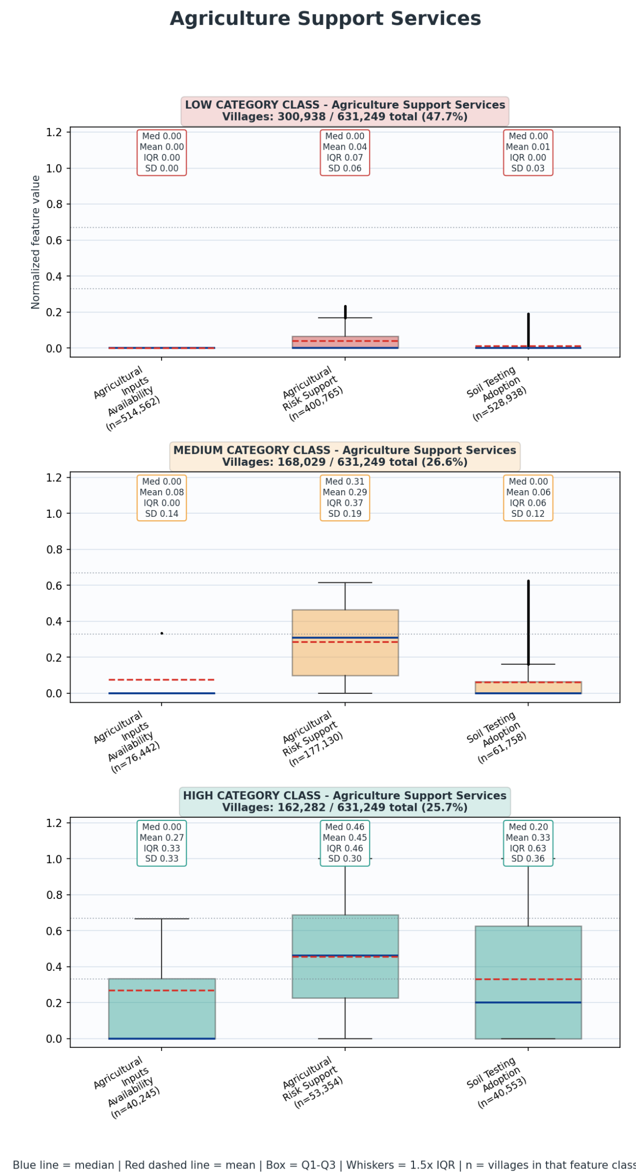

Figure 23: Agriculture Support Services

Our cluster results show that Agriculture Support Services are uneven and structurally underdeveloped. At the category level, **300,938 villages are Low, 168,029 are Medium, and 162,282 are High**, which appears moderately distributed. However, the feature-level structure reveals much sharper weakness: **Agricultural Inputs Availability is Low in 514,562 villages**, **Agricultural Risk Support is Low in 400,765 villages**, and **Soil Testing Adoption is Low in 528,938 villages**. This implies that even where some villages reach Medium or High at the composite category level, the underlying support environment is not uniformly strong. The broad development implication is that many villages may be agriculturally active without adequate access to formal input systems, risk protection, or evidence-based soil management. This is a serious concern because high cultivation without support services can intensify farmer vulnerability rather than reduce it.

LOW Agriculture Support Services villages

Low Agriculture Support Services villages should be understood as villages where farming is likely to occur without a sufficiently strong formal support architecture. These villages may lack nearby fertilizer shops, government seed centres, or soil testing centres; they may have limited crop insurance benefit coverage or weak registration under PM Kisan Pension Yojana; and farmers may not be using soil testing reports to guide fertilizer application. The developmental concern is not merely administrative absence but the social and economic consequences of unsupported cultivation. In such villages, farmers may rely on informal input dealers, inherited fertilizer practices, local knowledge without scientific validation, or private credit and risk-bearing mechanisms. This can lead to inefficient input use, soil degradation, low productivity, higher exposure to crop failure, and limited capacity to absorb climatic or market shocks. **Low support-service villages become especially important when they overlap with high Farm Employment or medium/high Agriculture Land Cultivation, because that combination indicates that many households depend on farming while the institutional support system around farming remains weak.** These villages should be prioritized for foundational service access: farmer advisory camps, soil testing drives, practical interpretation of soil reports, local access to quality seed and fertilizer, PMFBY enrolment and claim-support awareness, PMKPY registration facilitation, and linkage with credit and extension systems. The aim should be to build a basic farmer-support floor before expecting productivity gains, crop diversification, organic transition, or market-oriented agriculture.

MEDIUM Agriculture Support Services villages

They may have some risk-support coverage but weak input availability; some input access but poor soil-testing adoption; or some soil testing activity but little evidence that farmers are applying fertilizer according to soil reports. The analytical significance of the Medium cluster is that it points to partial institutional presence without full behavioural or livelihood conversion. In these villages, the development challenge is not only to expand services, but to convert available services into effective farmer practice. A soil testing report, for example, has limited value unless farmers understand it, trust it, and can access the recommended nutrients. Similarly, PMFBY registration or benefit receipt becomes meaningful only when farmers understand coverage, claim processes, and risk protection. Medium villages are therefore suited to a second-stage intervention strategy: not merely creating facilities, but strengthening uptake, interpretation, trust, and continuity. These villages can benefit from Panchayat-level agricultural service days, farmer field schools, demonstration plots, SHG/FPO-based advisory sessions, crop insurance facilitation, and convergence with irrigation, watershed, and market planning. They are also good candidates for testing more integrated agricultural planning because some support base already exists and can be deepened. Medium villages should be diagnosed by their weakest feature. The action required for weak input availability is different from the action required for weak risk support or low soil-testing adoption.

HIGH Agriculture Support Services villages

High Agriculture Support Services villages have comparatively stronger support ecosystems and should be treated as agricultural capability hubs. These villages are more likely to have better access to input institutions, stronger risk-support coverage, and higher soil-test-based fertilizer adoption. However, High status should not be interpreted as guaranteed prosperity or productivity. Support services are enabling conditions; their value depends on whether they are linked to water security, credit, producer institutions, storage, and markets. A High village with weak Agricultural Markets may produce better but still sell poorly. A

High village with weak Financial Inclusion may have technical advice but insufficient capital to act on it. A High village with weak Institutionalization may lack the collective capacity for input procurement, crop aggregation, insurance enrolment, or market negotiation. The policy value of High villages lies in their potential to support advanced agricultural strategies: soil-health-based nutrient management, crop diversification, water-use efficiency, climate-resilient farming, FPO/PACS strengthening, collective input procurement, organic or low-input pilots where suitable, and market-linked production planning. Where these villages also have strong Road Connectivity, Civic Infrastructure, and Institutionalization, they can function as service nodes for nearby villages with weaker support systems. High Agriculture Support Services villages should be used for consolidation and demonstration, but only after checking whether markets, credit, irrigation, and producer institutions are strong enough to convert support services into farmer income.

6.21 Agricultural Markets

The **Agricultural Markets** category combines two features: **Market Access** and **Food Storage Availability**. Market Access is scored from the type of market available: **mandi = 1.00, regular market = 0.75, weekly haat = 0.50, and no local market = 0.00**. Food Storage Availability is a binary feature based on whether a food-grain storage warehouse is available. The final category is calculated as the mean of the two feature cluster scores. This category therefore captures both **sales access** and **post-harvest holding capacity**. It is not only about whether farmers can produce; it is about whether they can sell, aggregate, store, and avoid distress sale. Agricultural Markets is one of the weakest structural categories. It has **449,486 villages in Low, 84,819 in Medium, and 96,944 in High**, equivalent to **71.2% Low, 13.4% Medium, and 15.4% High**.

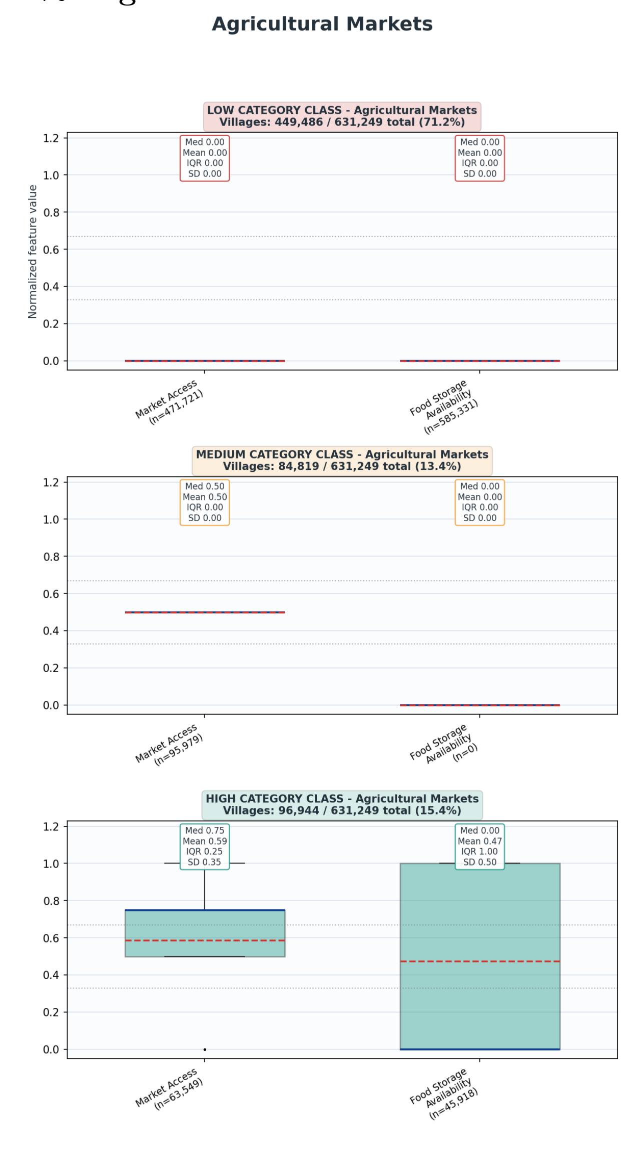

Figure 24: Agricultural Markets

Feature-level metrics show that both components are weak. **Market Access** is High in only 63,549. **Food Storage Availability** is High in only 45,918. Storage is especially rare. This is one of the most important findings in the entire analysis. Road Connectivity and Land Cultivation are relatively strong, but Agricultural Markets are weak. This means many villages may be physically connected and agriculturally active but still lack local market systems and storage.

LOW Agricultural Markets villages

Low Agricultural Markets villages are likely to lack mandi access, regular markets, weekly haats, or food-grain storage warehouses. These villages face serious post-production constraints. Farmers may have to sell immediately after harvest, travel farther to markets, depend on intermediaries, or accept lower prices due to lack of aggregation and storage. Low market access is especially concerning in villages with **High Land Cultivation**, **High Farm Employment**, or **High Irrigation and Watershed**. These villages may have strong production potential but weak sales infrastructure. Low Agricultural Markets also affects specialized livelihoods: fisheries, livestock, forest produce, cottage industry, organic farming, and alternative farming. Without markets and storage, these livelihood categories cannot scale.

MEDIUM Agricultural Markets villages

Medium villages may have partial market access, such as a weekly haat but no regular market or storage. They may also have some market access but weak storage. These villages are good candidates for market-system upgrading. The priority for Medium villages is to identify whether the main gap is market frequency, market quality, storage, aggregation, transport, or institutional linkage. A weekly haat may be enough for some products but not for high-volume agricultural produce, dairy, fish, or perishable vegetables. Storage gaps can be more important than market gaps in grain-producing villages. Medium villages with good **Road Connectivity** can often be upgraded through aggregation points, transport linkages, storage facilities, and buyer connections rather than requiring a full mandi in every village.

HIGH Agricultural Markets villages

High Agricultural Markets villages have relatively better market and/or storage access. These villages can act as rural market hubs for surrounding areas. They are important not only for their own farmers but for nearby villages that lack local market infrastructure. High market villages should be connected with **Institutionalization** and **Financial Inclusion**. FPOs, PACS, SHGs, producer groups, and credit systems can use these villages as aggregation and transaction points. High market villages with strong roads and energy can also support processing, storage, sorting, grading, packaging, cold-chain, and value addition. However, High Agricultural Markets should still be checked against **Agriculture Support Services**. Market access without production support may not produce high-quality, reliable output. Conversely, strong support services without markets may not improve incomes.

7 Conclusions

Overall, the cluster analysis shows that village development is not a single ladder but a set of interacting systems. Basic connectivity, agricultural land use, and some human development services have reached many villages, but institutional depth, financial inclusion, agricultural support, market access, water-sanitation systems, and livelihood diversification remain uneven. The most effective use of this analysis is therefore not only to identify Low villages category by category, but to identify combinations of deficits. Villages with multiple linked deficits should receive integrated intervention packages, while villages with strong enabling conditions should be used as growth nodes, demonstration sites, and service hubs.

8 Appendix: Calculations, Column Usage, Feature Creation, Category Creation

Table 3: Calculations, Column Usage, Feature Creation, and Category Creation

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Institutionalization	shg_pen_feature	total_hhd; total_hhd_mobilized_into_shg		CALC::shg_pen_feature := normalize(total_hhd_mobilized_into_shg / total_hhd)
Institutionalization	shg_fed_feature	total_no_of_shg_promoted; total_shg		CALC::shg_fed_feature := normalize(total_no_of_shg_promoted / total_shg)
Institutionalization	pg_pen_feature	total_hhd; total_hhd_mobilized_into_pg		CALC::pg_pen_feature := normalize(total_hhd_mobilized_into_pg / total_hhd)
Institutionalization	fpo_feature	availability_of_fpos_pacs	farmers_collective_score	CALC::farmers_collective_score := ordinal_map(availability_of_fpos_pacs); FORMULA::FPO=1, PACS=1, both FPO and PACS=2, none=0; normalized during feature scoring CALC::fpo_feature := normalize(farmers_collective_score)
Institutionalization	institutionalization_category			CALC::institutionalization_category := mean(feature_class_score(shg_pen_feature, shg_fed_feature, pg_pen_feature, fpo_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Social Protection	pds_util_feature	gp_total_hhd_eligible_under_nfsa; gp_total_hhd_receiving_food_grains_from_fps		CALC::pds_util_feature := normalize(gp_total_hhd_receiving_food_grains_from_fps / gp_total_hhd_eligible_under_nfsa)
Social Protection	bpl_cov_feature	total_hhd; total_hhd_having_bpl_cards		CALC::bpl_cov_feature := normalize(total_hhd_having_bpl_cards / total_hhd)
Social Protection	nfsa_cov_feature	gp_total_hhd_eligible_under_nfsa; total_hhd		CALC::nfsa_cov_feature := normalize(gp_total_hhd_eligible_under_nfsa / total_hhd)
Social Protection	pension_cov_feature	total_hhd; total_hhd_availing_pension_under_nsap		CALC::pension_cov_feature := normalize(total_hhd_availing_pension_under_nsap / total_hhd)
Social Protection	social_protection_category			CALC::social_protection_category := mean(feature_class_score(pds_util_feature, bpl_cov_feature, nfsa_cov_feature, pension_cov_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Civic Infrastructure	panchayat_bhawan_feature	availability_of_panchayat_bhawan		CALC::panchayat_bhawan_feature := binary_presence(availability_of_panchayat_bhawan)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Civic Infrastructure	post_office_feature	is_post_office_available		CALC::post_office_feature := binary_presence(is_post_office_available)
Civic Infrastructure	rep_training_feature	total_no_of_elect_rep_undergone_training_under_rgsa; total_no_of_elected_representatives		CALC::rep_training_feature := normalize(total_no_of_elect_rep_undergone_training_under_rgsa / total_no_of_elected_representatives)
Civic Infrastructure	rep_orientation_feature	total_no_of_elect_rep_oriented_under_rgsa; total_no_of_elected_representatives		CALC::rep_orientation_feature := normalize(total_no_of_elect_rep_oriented_under_rgsa / total_no_of_elected_representatives)
Civic Infrastructure	info_board_feature	availability_of_public_information_board		CALC::info_board_feature := binary_presence(availability_of_public_information_board)
Civic Infrastructure	public_library_feature	availability_of_public_library		CALC::public_library_feature := binary_presence(availability_of_public_library)
Civic Infrastructure	civic_infrastructure_category			CALC::civic_infrastructure_category := mean(feature_class_score(panchayat_bhawan_feature, post_office_feature, rep_training_feature, rep_orientation_feature, info_board_feature, public_library_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Financial Inclusion	bank_feature	is_bank_available		CALC::bank_feature := binary_presence(is_bank_available)
Financial Inclusion	atm_feature	is_atm_available		CALC::atm_feature := binary_presence(is_atm_available)
Financial Inclusion	bank_correspondent_feature	is_bank_buss_correspondent_with_internet		CALC::bank_correspondent_feature := binary_presence(is_bank_buss_correspondent_with_internet)
Financial Inclusion	shg_credit_feature	total_shg; total_shg_accessed_bank_loans		CALC::shg_credit_feature := normalize(total_shg_accessed_bank_loans / total_shg)
Financial Inclusion	jan_dhan_pen_feature	total_hhd; total_hhd_availing_pmjdy_bank_ac		CALC::jan_dhan_pen_feature := normalize(total_hhd_availing_pmjdy_bank_ac / total_hhd)
Financial Inclusion	financial_inclusion_category			CALC::financial_inclusion_category := mean(feature_class_score(bank_feature, atm_feature, bank_correspondent_feature, shg_credit_feature, jan_dhan_pen_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Energy Access	electrification_rate_feature	availablility_hours_of_domestic_electricity	domestic_electricity_hours_score	CALC::domestic_electricity_hours_score := ordinal_map(availablility_hours_of_domestic_electricity); FORMULA::1-4 hrs=0.25, 4-8 hrs=0.50, 8-12 hrs=0.75, >12 hrs=1.00, no electricity=0.00 CALC::electrification_rate_feature := normalize(domestic_electricity_hours_score)
Energy Access	electricity_supply_to_msme_feature	availability_of_elect_supply_to_msme		CALC::electricity_supply_to_msme_feature := binary_presence(availability_of_elect_supply_to_msme)
Energy Access	clean_energy_penetration_feature	total_hhd; total_hhd_with_clean_energy		CALC::clean_energy_penetration_feature := normalize(total_hhd_with_clean_energy / total_hhd)
Energy Access	energy_access_category			CALC::energy_access_category := mean(feature_class_score(electrification_rate_feature, electricity_supply_to_msme_feature, clean_energy_penetration_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Road Connectivity	all_weather_road_feature	is_village_connected_to_all_weather_road		CALC::all_weather_road_feature := binary_presence(is_village_connected_to_all_weather_road)
Road Connectivity	internal_pucca_road_feature	availability_of_internal_pucca_road	internal_pucca_road_quality_score	CALC::internal_pucca_road_quality_score := ordinal_map(availability_of_internal_pucca_road); FORMULA::fully covered=1.00, partially covered=0.50, not covered=0.00 CALC::internal_pucca_road_feature := normalize(internal_pucca_road_quality_score)
Road Connectivity	public_transport_feature	availability_of_public_transport	public_transport	CALC::public_transport_feature := normalize(public_transport)
Road Connectivity	railway_station_feature	availability_of_railway_station		CALC::railway_station_feature := binary_presence(availability_of_railway_station)
Road Connectivity	road_connectivity_category			CALC::road_connectivity_category := mean(feature_class_score(all_weather_road_feature, internal_pucca_road_feature, public_transport_feature, railway_station_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Housing Quality	pucca_housing_rate_feature	total_hhd; total_hhd_with_kuccha_wall_kuccha_roof	pucca_housing_rate_score	CALC::pucca_housing_rate_score := 1.0 - min(1.0, total_hhd_with_kuccha_wall_kuccha_roof / total_hhd); INVERSE::higher score means lower burden/rate CALC::pucca_housing_rate_feature := normalize(pucca_housing_rate_score)

category	feature__id	raw__columns__used	derived__variables	calculations__formulae
Housing Quality	housing__scheme__coverage__feature	total__hhd; total__hhd__got__benefit__under__state__housing__scheme; total__hhd__have__got__pmay__house	housing__scheme__coverage__score	CALC::housing__scheme__coverage__score := weighted__mean(pmay__and__state__housing__benefit__share); FORMULA::min(1.0, (total__hhd__have__got__pmay__house + total__hhd__got__benefit__under__state__housing__scheme) / total__hhd) CALC::housing__scheme__coverage__feature := normalize(housing__scheme__coverage__score)
Housing Quality	pmay__demand__met__feature	total__hhd__have__got__pmay__house; total__hhd__in__pmay__permanent__wait__list	pmay__demand__met__score	CALC::pmay__demand__met__score := weighted__mean(pmay__houses__over__pmay__houses__plus__waitlist); FORMULA::min(1.0, total__hhd__have__got__pmay__house / (total__hhd__have__got__pmay__house + total__hhd__in__pmay__permanent__wait__list)) CALC::pmay__demand__met__feature := normalize(pmay__demand__met__score)
Housing Quality	ujjwala__coverage__feature	total__hhd; total__hhd__availing__pmuy__benefits	ujjwala__coverage__score	CALC::ujjwala__coverage__score := min(1.0, total__hhd__availing__pmuy__benefits / total__hhd) CALC::ujjwala__coverage__feature := normalize(ujjwala__coverage__score)
Housing Quality	housing__quality__category			CALC::housing__quality__category := mean(feature__class__score(pucca__housing__rate__feature, housing__scheme__coverage__feature, pmay__demand__met__feature, ujjwala__coverage__feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL__CLASS::MiniBatchKMeans category classification

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Maternal and Child Health	awc_infra_enrollment_coverage_feature	availability_of_mother_child_health_facilities; is_aanganwadi_centre_available; is_early_childhood_edu_provided_in_anganwadi; total_childs_aged_0_to_3_years; total_childs_aged_0_to_3_years_reg_under_aanganwadi; total_childs_aged_3_to_6_years_reg_under_aanganwadi; total_female_child_age_bw_0_6; total_male_child_age_bw_0_6; total_no_of_children_in_icds_cas; total_no_of_registered_children_in_anganwadi	awc_0_3_coverage_score; awc_3_6_coverage_score; awc_infra_enrollment_coverage_score; awc_total_enrollment_score; children_3_6_estimated; children_under_age_6; overall_awc_enrollment_score	CALC::awc_0_3_coverage_score := min(1.0, total_childs_aged_0_to_3_years_reg_under_aanganwadi / total_childs_aged_0_to_3_years) CALC::children_under_age_6 := total_male_child_age_bw_0_6 + total_female_child_age_bw_0_6 CALC::children_3_6_estimated := max(0, children_under_age_6 - total_childs_aged_0_to_3_years) CALC::awc_3_6_coverage_score := min(1.0, total_childs_aged_3_to_6_years_reg_under_aanganwadi / children_3_6_estimated) CALC::overall_awc_enrollment_score := min(1.0, total_no_of_children_in_icds_cas / children_under_age_6) CALC::awc_total_enrollment_score := min(1.0, total_no_of_registered_children_in_anganwadi / children_under_age_6) CALC::awc_infra_enrollment_coverage_score := mean(awc_available, mch_facility_available, ece_available, awc_0_3_coverage, awc_3_6_coverage, overall_awc_enrollment, awc_total_enrollment); FORMULA::mean(awc_available, mch_facility_available, ece_available, awc_0_3_coverage_score, awc_3_6_coverage_score, overall_awc_enrollment_score, awc_total_enrollment_score) CALC::awc_infra_enrollment_coverage_feature := normalize(awc_infra_enrollment_coverage_score)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Maternal and Child Health	maternal_health_care_access_feature	total_anemic_pregnant_women; total_no_of_lactating_mothers; total_no_of_lactating_mothers_receiving_services_under_icds; total_no_of_pregnant_women; total_no_of_pregnant_women_receiving_services_under_icds; total_no_of_women_delivered_babies_at_hospitals_registered_asha	institutional_delivery_rate_score; lactating_service_coverage_score; maternal_anemia_score; maternal_health_care_access_score; pregnant_service_coverage_score	CALC::pregnant_service_coverage_score := min(1.0, total_no_of_pregnant_women_receiving_services_under_icds / total_no_of_pregnant_women) CALC::lactating_service_coverage_score := min(1.0, total_no_of_lactating_mothers_receiving_services_under_icds / total_no_of_lactating_mothers) CALC::maternal_anemia_score := 1.0 - min(1.0, total_anemic_pregnant_women / total_no_of_pregnant_women); INVERSE::higher score means lower burden/rate CALC::institutional_delivery_rate_score := min(1.0, total_no_of_women_delivered_babies_at_hospitals_registered_asha / total_no_of_pregnant_women) CALC::maternal_health_care_access_score := mean(pregnant_service_coverage, lactating_service_coverage, maternal_anemia_score, institutional_delivery_rate); FORMULA::mean(pregnant_service_coverage_score, lactating_service_coverage_score, maternal_anemia_score, institutional_delivery_rate_score) CALC::maternal_health_care_access_feature := normalize(maternal_health_care_access_score)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Maternal and Child Health	child_nutrition_development_feature	total_childs_aged_0_to_3_years; total_childs_aged_0_to_3_years_immunized; total_childs_categorized_non_stunted_as_per_icds; total_female_child_age_bw_0_6; total_male_child_age_bw_0_6; total_no_of_children_in_icds_cas; total_no_of_registered_children_in_anganwadi; total_no_of_young_anemic_children_6_59_months_in_icds_cas; total_underweight_child_age_under_6_years	child_anemia_score; child_immunization_rate_score; child_nutrition_development_score; child_nutrition_score; child_stunting_score; children_under_age_6; stunted_children	CALC::child_immunization_rate_score := min(1.0, total_childs_aged_0_to_3_years_immunized / total_childs_aged_0_to_3_years) CALC::children_under_age_6 := total_male_child_age_bw_0_6 + total_female_child_age_bw_0_6 CALC::child_nutrition_score := 1.0 - min(1.0, total_underweight_child_age_under_6_years / children_under_age_6); INVERSE::higher score means lower burden/rate CALC::stunted_children := max(0, total_no_of_registered_children_in_anganwadi - total_childs_categorized_non_stunted_as_per_icds) CALC::child_stunting_score := 1.0 - min(1.0, stunted_children / total_no_of_registered_children_in_anganwadi); INVERSE::higher score means lower burden/rate CALC::child_anemia_score := 1.0 - min(1.0, total_no_of_young_anemic_children_6_59_months_in_icds_cas / total_no_of_children_in_icds_cas); INVERSE::higher score means lower burden/rate; NULL_RULE::denominator<=0 is treated as NULL before feature scoring CALC::child_nutrition_development_score := mean(child_immunization_rate, child_nutrition_score, child_stunting_score, child_anemia_score); FORMULA::mean(child_immunization_rate_score, child_nutrition_score, child_stunting_score, child_anemia_score) CALC::child_nutrition_development_feature := normalize(child_nutrition_development_score)
Maternal and Child Health	newborn_health_outcomes_feature	total_no_of_newly_born_children; total_no_of_newly_born_underweight_children	newborn_health_score	CALC::newborn_health_score := 1.0 - min(1.0, total_no_of_newly_born_underweight_children / total_no_of_newly_born_children); INVERSE::higher score means lower burden/rate; NULL_RULE::denominator<=0 is treated as NULL before feature scoring CALC::newborn_health_outcomes_feature := normalize(newborn_health_score)

category	feature__id	raw__columns__used	derived__variables	calculations__formulae
Maternal and Child Health	health__schemes__utilization__feature	gp__total__no__of__beneficiaries__receiving__benefits__under__pmjay; gp__total__no__of__eligible__beneficiaries__under__pmjay; total__hhd; total__hhd__registered__under__pmjay; total__no__of__beneficiaries__receiving__benefits__under__pmmvy; total__no__of__eligible__beneficiaries__under__pmmvy	ayushman__bharat__utilization__score; health__insurance__coverage__score; health__schemes__utilization__score; matru__vandana__utilization__score	CALC::matru__vandana__utilization__score := min(1.0, total__no__of__beneficiaries__receiving__benefits__under__pmmvy / total__no__of__eligible__beneficiaries__under__pmmvy) CALC::health__insurance__coverage__score := min(1.0, total__hhd__registered__under__pmjay / total__hhd) CALC::ayushman__bharat__utilization__score := min(1.0, gp__total__no__of__beneficiaries__receiving__benefits__under__pmjay / gp__total__no__of__eligible__beneficiaries__under__pmjay) CALC::health__schemes__utilization__score := mean(matru__vandana__utilization, health__insurance__coverage, ayushman__bharat__utilization); FORMULA::mean(matru__vandana__utilization__score, health__insurance__coverage__score, ayushman__bharat__utilization__score) CALC::health__schemes__utilization__feature := normalize(health__schemes__utilization__score)
Maternal and Child Health	maternal__child__health__category			CALC::maternal__child__health__category := mean(feature__class__score(awc__infra__enrollment__coverage__feature, maternal__health__care__access__feature, child__nutrition__development__feature, newborn__health__outcomes__feature, health__schemes__utilization__feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Water and Sanitation	piped__water__coverage__feature	availability__of__piped__tap__water; total__hhd; total__hhd__having__piped__water__connection	piped__tap__water__coverage__score; piped__water__coverage__score	CALC::piped__tap__water__coverage__score := ordinal__map(availability__of__piped__tap__water); FORMULA::100% habitations=1.00, 50-100%=0.60, <50%=0.25, only one habitation=0.10, not covered=0.00 CALC::piped__water__coverage__score := mean(habitation__coverage, household__connection__ratio); FORMULA::mean(piped__tap__water__coverage__score, clipped(total__hhd__having__piped__water__connection / total__hhd)) CALC::piped__water__coverage__feature := normalize(piped__water__coverage__score)
Water and Sanitation	sanitation__coverage__feature	total__hhd; total__hhd__not__having__sanitary__latrines		CALC::sanitation__coverage__feature := normalize(total__hhd__not__having__sanitary__latrines / total__hhd); INVERSE::1 - normalized ratio

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Water and Sanitation	drainage_quality_feature	availability_of_drainage_system	drainage_quality_score	CALC::drainage_quality_score := ordinal_map(availability_of_drainage_system); FORMULA::closed=1.00, covered open=0.75, uncovered open=0.50, kuchha=0.25, no drainage=0.00 CALC::drainage_quality_feature := normalize(drainage_quality_score)
Water and Sanitation	waste_disposal_feature	is_community_waste_disposal_system		CALC::waste_disposal_feature := binary_presence(is_community_waste_disposal_system)
Water and Sanitation	biogas_waste_recycling_feature	is_community_biogas_waste_recycle_for_production		CALC::biogas_waste_recycling_feature := binary_presence(is_community_biogas_waste_recycle_for_production)
Water and Sanitation	water_sanitation_category			CALC::water_sanitation_category := mean(feature_class_score(piped_water_coverage_feature, sanitation_coverage_feature, drainage_quality_feature, waste_disposal_feature, biogas_waste_recycling_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Livelihoods Cottage and Traditional Industry	cottage_units_available_feature	availability_of_cottage_small_scale_units		CALC::cottage_units_available_feature := binary_presence(availability_of_cottage_small_scale_units)
Livelihoods Cottage and Traditional Industry	cottage_industry_participation_feature	total_hhd; total_hhd_engaged_cottage_small_scale_units		CALC::cottage_industry_participation_feature := normalize(total_hhd_engaged_cottage_small_scale_units / total_hhd)
Livelihoods Cottage and Traditional Industry	handloom_feature	is_handloom		CALC::handloom_feature := binary_presence(is_handloom)
Livelihoods Cottage and Traditional Industry	handicrafts_feature	is_handicrafts		CALC::handicrafts_feature := binary_presence(is_handicrafts)
Livelihoods Cottage and Traditional Industry	livelihoods_cottage_traditional_industry_category			CALC::livelihoods_cottage_traditional_industry_category := mean(feature_class_score(cottage_units_available_feature, cottage_industry_participation_feature, handloom_feature, handicrafts_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Farm Employment	farm_employment_feature	total_hhd; total_hhd_engaged_in_farm_activities		CALC::farm_employment_feature := normalize(total_hhd_engaged_in_farm_activities / total_hhd)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Farm Employment	livelihoods_employment_category			CALC::livelihoods_employment_category := mean(feature_class_score(farm_employment_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Livelihoods Forest Resources	community_forest_feature	availability_of_community_forest		CALC::community_forest_feature := binary_presence(availability_of_community_forest)
Livelihoods Forest Resources	minor_forest_production_feature	availability_of_minor_forest_production		CALC::minor_forest_production_feature := binary_presence(availability_of_minor_forest_production)
Livelihoods Forest Resources	forest_dependence_feature	total_hhd; total_hhd_source_of_minor_forest_production		CALC::forest_dependence_feature := normalize(total_hhd_source_of_minor_forest_production / total_hhd)
Livelihoods Forest Resources	livelihoods_forest_resources_category			CALC::livelihoods_forest_resources_category := mean(feature_class_score(community_forest_feature, minor_forest_production_feature, forest_dependence_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Livelihoods Common Resources	common_pastures_feature	is_common_pastures_available		CALC::common_pastures_feature := binary_presence(is_common_pastures_available)
Livelihoods Common Resources	livelihoods_common_resources_category			CALC::livelihoods_common_resources_category := mean(feature_class_score(common_pastures_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Livelihoods Alternative Farming	alternative_farming_feature	is_bee_farming; is_sericulture		CALC::alternative_farming_feature := mean(binary_presence(is_bee_farming, is_sericulture))
Livelihoods Alternative Farming	livelihoods_alternative_farming_category			CALC::livelihoods_alternative_farming_category := mean(feature_class_score(alternative_farming_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Livelihoods Fisheries	fisheries_aquaculture_feature	availability_of_aquaculture_ext_facility; availability_of_fish_community_ponds; availability_of_fish_farming	fisheries_aquaculture_score	CALC::fisheries_aquaculture_score := mean(fish_farming_available, community_ponds_available, aquaculture_extension_facility_available); FORMULA::mean(fish_farming_available, fish_community_pond_available, aquaculture_extension_facility_available) CALC::fisheries_aquaculture_feature := normalize(fisheries_aquaculture_score)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Livelihoods Fisheries	livelihoods_fisheries_category			CALC::livelihoods_fisheries_category := mean(feature_class_score(fisheries_aquaculture_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Livestock and Veterinary	veterinary_services_feature	availability_of_livestock_extension_services; is_veterinary_hospital_available	livestock_extension_services_score; veterinary_services_access_score	CALC::livestock_extension_services_score := ordinal_map(availability_of_livestock_extension_services); FORMULA::1 if availability_of_livestock_extension_services is 1 or 2, else 0 CALC::veterinary_services_access_score := mean(veterinary_hospital_available, livestock_extension_service_access); FORMULA::mean(veterinary_hospital_available, livestock_extension_services_score) CALC::veterinary_services_feature := normalize(veterinary_services_access_score)
Livestock and Veterinary	development_projects_feature	availability_of_goatary_dev_project; availability_of_pigery_development; availability_of_poultry_dev_project		CALC::development_projects_feature := mean(binary_presence(availability_of_poultry_dev_project, availability_of_goatary_dev_project, availability_of_pigery_development))
Livestock and Veterinary	milk_routes_feature	availability_of_milk_routes		CALC::milk_routes_feature := binary_presence(availability_of_milk_routes)
Livestock and Veterinary	livestock_veterinary_category			CALC::livestock_veterinary_category := mean(feature_class_score(veterinary_services_feature, development_projects_feature, milk_routes_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Agriculture Land Cultivation	land_utilization_feature	area_irrigated_in_hac; net_sown_area_in_hac; net_sown_area_kharif_in_hac; net_sown_area_other_in_hac; net_sown_area_rabi_in_hac; total_cultivable_area_in_hac	cropping_intensity; land_utilization_feature_score	CALC::cropping_intensity := weighted_mean(weighted_seasonal_sown_area_over_net_sown_area); FORMULA::min((kharif + (2 * rabi) + (3 * other)) / net_sown, 3) / 3 CALC::land_utilization_feature_score := mean(land_cultivation_rate, irrigation_coverage, cropping_intensity); FORMULA::mean(clipped(net_sown_area_in_hac / total_cultivable_area_in_hac), clipped(area_irrigated_in_hac / total_cultivable_area_in_hac), cropping_intensity) CALC::land_utilization_feature := normalize(land_utilization_feature_score)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Agriculture Land Cultivation	agriculture_land_cultivation_category			CALC::agriculture_land_cultivation_category := mean(feature_class_score(land_utilization_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Agriculture Irrigation and Watershed	irrigation_infra_watershed_dev_feature	availability_of_major_source_of_irrigation; availability_of_rain_harvest_system; availability_of_watershed_dev_project	canal	CALC::irrigation_infra_watershed_dev_feature := mean(binary_presence(canal, availability_of_rain_harvest_system, availability_of_watershed_dev_project))
Agriculture Irrigation and Watershed	nrega_nrm_exp_feature	total_approved_labour_budget_for_year; total_expenditure_approved_under_nrm_labour_budget_during_yr		CALC::nrega_nrm_exp_feature := normalize(total_expenditure_approved_under_nrm_labour_budget_during_yr / total_approved_labour_budget_for_year)
Agriculture Irrigation and Watershed	modern_irrigation_feature	no_of_farmers_using_drip_sprinkler; total_no_of_farmers		CALC::modern_irrigation_feature := normalize(no_of_farmers_using_drip_sprinkler / total_no_of_farmers)
Agriculture Irrigation and Watershed	agriculture_irrigation_watershed_category			CALC::agriculture_irrigation_watershed_category := mean(feature_class_score(irrigation_infra_watershed_dev_feature, nrega_nrm_exp_feature, modern_irrigation_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Agriculture Organic Farming	organic_farming_feature	total_no_farmers_adopted_organic_farming; total_no_of_farmers		CALC::organic_farming_feature := normalize(total_no_farmers_adopted_organic_farming / total_no_of_farmers)
Agriculture Organic Farming	agriculture_organic_farming_category			CALC::agriculture_organic_farming_category := mean(feature_class_score(organic_farming_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Agriculture Support Services	agri_inputs_availability_feature	is_fertilizer_shop_available; is_govt_seed_centre_available; is_soil_testing_centre_available	agri_inputs_availability_score	CALC::agri_inputs_availability_score := mean(soil_testing_centre_available, fertilizer_shop_available, government_seed_centre_available); FORMULA::mean(soil_testing_centre_available, fertilizer_shop_available, government_seed_centre_available) CALC::agri_inputs_availability_feature := normalize(agri_inputs_availability_score)

category	feature_id	raw_columns_used	derived_variables	calculations_formulae
Agriculture Support Services	agri_risk_support_feature	total_no_of_farmers; total_no_of_farmers_received_benefit_under_pmfby; total_no_of_farmers_registered_under_pmkpy	agri_risk_support_score	CALC::agri_risk_support_score := mean(pmkpy_registration_coverage, crop_insurance_coverage); FORMULA::mean(clipped(total_no_of_farmers_registered_under_pmkpy / total_no_of_farmers), clipped(total_no_of_farmers_received_benefit_under_pmfby / total_no_of_farmers)) CALC::agri_risk_support_feature := normalize(agri_risk_support_score)
Agriculture Support Services	soil_testing_adoption_feature	total_no_of_farmers; total_no_of_farmers_add_fert_in_soil_as_per_report		CALC::soil_testing_adoption_feature := normalize(total_no_of_farmers_add_fert_in_soil_as_per_report / total_no_of_farmers)
Agriculture Support Services	agriculture_support_services_category			CALC::agriculture_support_services_category := mean(feature_class_score(agri_inputs_availability_feature, agri_risk_support_feature, soil_testing_adoption_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification
Agricultural Markets	market_access_feature	availability_of_market	market_access_score	CALC::market_access_score := ordinal_map(availability_of_market); FORMULA::mandi=1.00, regular market=0.75, weekly haat=0.50, no local market=0.00 CALC::market_access_feature := normalize(market_access_score)
Agricultural Markets	food_storage_feature	availability_of_food_storage_warehouse		CALC::food_storage_feature := binary_presence(availability_of_food_storage_warehouse)
Agricultural Markets	agricultural_markets_category			CALC::agricultural_markets_category := mean(feature_class_score(market_access_feature, food_storage_feature)); CLASS_SCORE::Low=0, Medium=0.5, High=1; FINAL_CLASS::MiniBatchKMeans category classification